



Coupled EISPH–OSPD framework for modeling hydroelastic response and brittle fracture

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Received: 28 May 2025 / Accepted: 10 September 2025
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Abstract

Accurate simulation of fluid–structure interaction (FSI) involving incompressible flows and deformable solids remains a significant computational challenge due to the nonlinear coupling and complex interface dynamics. This study presents a coupling framework integrating the Explicit Incompressible Smoothed Particle Hydrodynamics (EISPH) method for fluid modeling with the Ordinary State-Based Peridynamics (OSPD) formulation for solid mechanics. The framework aims to capture FSI phenomena encompassing large elastic deformations and structural failures. The elastic response of the OSPD model is validated through cantilever beam simulations, highlighting the efficacy of surface correction techniques and second-order predictor–corrector time integration schemes. Fracture behavior is assessed via the Kalthoff–Winkler experiment, demonstrating the model’s capability in simulating crack propagation. The coupling strategy employs momentum exchange and subcycling-based time integration to ensure numerical stability and physical fidelity. Benchmark tests, including dam-break scenarios and free-surface flow interactions with elastic structures, confirm the robustness and accuracy of the framework. Notably, the incompressible SPH method contributes to pressure stability without necessitating additional stabilization terms, even in simulations involving structural failure. Overall, the developed framework offers a physically consistent, stable, and accurate approach for simulating complex FSI problems.

Keywords Particle-method · EISPH · OSPD · Hydroelastic · Fracture · Fluid-structure interaction (FSI)

Nomenclature

$\mathbf{u}$	Velocity field	Δ_{SPH}	SPH initial particle spacing
ρ	Particle density	h	SPH smoothing length
m	Particle mass	$\delta \mathbf{u}$	SPH particle shifting velocity
V	Particle volume	$\mathbf{L}$	SPH renormalization matrix
P	Pressure	$\mathbf{n}$	SPH free-surface normal vector
μ	Dynamic viscosity	$\mathbf{u}$	Displacement field
$\mathbf{g}$	Gravitational acceleration	θ	Dilatation
i	SPH center particle	W	Strain energy density (SED)
j	SPH neighboring particle	G_0	Energy release
$\mathbf{r}_i, \mathbf{r}_j$	SPH position vector	K	Bulk modulus
Ω	SPH domain of interest	E	Young’s modulus
W_{ij}	SPH kernel	ν	Poisson ratio
		A	PD center particle
		$\mathbf{x}_A, \mathbf{x}_B$	PD reference position
		$\mathbf{y}_A, \mathbf{y}_B$	PD deformed position vector
		s_{AB}	PD stretch
		$\mathcal{H}$	PD domain of interest
		ω_{AB}	PD influence function
		Δ_{PD}	PD initial particle spacing
		$\mathbf{t}$	PD force state vector
		$\mathbf{g}$	PD correction factor

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1 Introduction

Fluid–structure interaction (FSI) phenomena are prevalent across a wide spectrum of scientific and engineering application [1–4]. A precise and comprehensive understanding of FSI systems is critical for the reliable design and analysis of such coupled systems. The inherently nonlinear coupling between fluid motion and structural deformation poses substantial computational challenges, particularly in cases involving large elastic deformations and complex interface dynamics. Although significant progress has been made in computational FSI modeling, achieving both numerical stability and physical fidelity, especially for problems involving incompressible flows interacting with deformable solids remains an open and active area of research.

Meshfree methods, such as Smoothed Particle Hydrodynamics (SPH; [5, 6]) and Moving Particle Semi-implicit (MPS; [7]), have been widely employed to tackle complex nonlinear FSI problems. SPH, in particular, has gained widespread adoption for simulating incompressible flows, free-surface dynamics, and multiphase interactions due to its Lagrangian nature and ease implementation [8–10]. Leveraging these advantages, numerous studies have applied SPH for modeling both fluid and structural aspects of FSI problems [11–14]. Furthermore, significant efforts have been made to advance the applicability and versatility of meshfree methods in complex FSI simulations. Particular attention has been given to the coupling of SPH with structural models based on mesh-based methods such as Finite Element Method (FEM) [15–17], and Smoothed Point Interpolation Method (SPIM) [18], as well as with other meshfree techniques, including the Discrete Element Method (DEM) [19], Lattice Particle Method (LPM) [20], and Peridynamics (PD) [21, 22].

Peridynamics (PD), a nonlocal continuum theory for solid mechanics, presents its strengths in modeling both discontinuous deformation and crack propagation [23–25]. Developed initially to handle brittle fracture, PD replaces spatial derivatives with integral expressions, offering enhanced robustness for simulating wave propagation and stress evolution in dynamically loaded solids [23]. Several PD formulations have been proposed to improve accuracy and computational efficiency, including Bond Based PD (BBPD), Ordinary State Based PD (OSPD), Non Ordinary State Based PD (NOSPD), and Dual Horizon PD [23, 24, 26]. Among those methods, BBPD offers simplicity but is limited by its fixed Poisson's ratio, making it suitable primarily for brittle fractures. OSPD removes this limitation and supports general linear elastic responses while retaining computational efficiency and mesh-free advantages. However, both BBPD and OSPD, being mostly limited to linear elastic formulations, face challenges in modeling finite

deformations or nonlinear material behavior. To address these challenges, NOSPD formulations have been introduced [27]. By incorporating deformation gradient and stress measures from classical continuum mechanics, NOSPD enables robust extensions to hypo-elastic/plastic [28], hyper-elastic/plastic [29], and even elastoplastic constitutive models [30].

Given the complementary strengths of SPH and PD, their coupling presents a promising approach for addressing a wide range of complex FSI problems, particularly those involving structural failure [31]. The first trial on SPH-PD studies have investigated the simulation of soil deformation and fragmentation induced by explosive loads using NOSPD coupled with compressible SPH [32–35]. More recently, increasing number of works have also focused on modeling FSI problems involving free surface flows interacting with structures composed of materials ranging from elastic to brittle. To this end, coupling strategies have been developed that integrate various PD formulations such as BBPD [21, 36], OSPD [22, 31, 37] and NOSPD [38, 39] with weakly compressible SPH schemes.

Since the original development of PD focused primarily on fracture modeling, comparatively less attention has been given to validating its performance on capturing its elastic behavior. To address this gap, a recent coupling of NOSPD with δ -SPH + has been proposed and thoroughly validated through simulations of the elastic response of solid bars, demonstrating the potential of this hybrid approach to accurately capture both fluid and structural dynamics in elastic conditions [39]. In the same study, significant pressure oscillations were observed when structural failure was incorporated using conventional WCSPH formulations. To mitigate these acoustic artifacts, an acoustic damping term, which has shown good performance in reducing pressure noise in standard WCSPH methods was introduced [40].

To the best of our knowledge, despite the growing interest in Incompressible SPH (ISPH) methods, which are inherently free from acoustic noise, no studies have yet been reported that couple ISPH with PD. In this work, we propose a novel coupling framework that integrates EISPH for fluid modeling, with OSPD chosen as a reasonable option for the solid phase due to its balance between modeling capability and computational simplicity, aiming to capture FSI phenomena involving large elastic deformations and structural failure. The EISPH method enforces incompressibility through local pressure projection, eliminating the need for global matrix solves and producing stable, noise-free pressure fields, thereby making it well suited for impact- and splash-dominated flows. OSPD, in turn, offers a fully mesh-free and non-local solid mechanics framework that naturally accommodates discontinuities and large deformations, while retaining computational simplicity and efficiency.

In **Sect. 2**, the elastic response of the OSPD formulation is rigorously validated using cantilever beam simulations, with a focus on time integration schemes, influence functions, and correction techniques. In addition, its fracture behavior is assessed through validation against the Kalthoff–Winkler experiment. With the OSPD model validated for both elastic and fracture behavior, and the SPH method validated using conventional dam-break problems involving pressure-driven responses, a robust coupling strategy is proposed in **Sect. 3**, addressing both physical boundary conditions and computational coupling schemes. In **Sect. 4**, the performance of the proposed framework is demonstrated through three representative FSI problems involving complex free-surface dynamics and large deformation of elastic bars under hydrodynamic pressure loads. Finally, to showcase the versatility and applicability of the framework, an FSI problem involving structural failure is simulated and validated, confirming its ability to accurately capture highly nonlinear coupled fluid–structure interactions under extreme conditions. Comprehensive summary and conclusions have been made in **Sect. 5**.

2 Numerical discretization of governing equations

In this study, the EISPH method is employed to model the fluid flow, while the OSPD approach is used to simulate the structural dynamics. This section outlines the fundamental formulation of each method and presents the discretized forms of the governing equations. Additionally, numerical treatments such as boundary condition implementation are described.

2.1 Governing equation for fluid flow

2.1.1 Governing equations in continuous form

The governing equations for isothermal, incompressible, and Newtonian fluid flow are based on the principles of mass conservation and momentum conservation, given as follows:

$$\nabla \cdot \mathbf{u} = 0, \quad (1)$$

$$\frac{D\mathbf{u}}{Dt} = -\frac{\nabla P}{\rho} + \frac{\mu}{\rho} \nabla^2 \mathbf{u} + \mathbf{g}, \quad (2)$$

where $\mathbf{u}$ represents the velocity field, ρ is the fluid density, P denotes the pressure, μ is the dynamic viscosity, $\mathbf{g}$ represents gravitational acceleration.

2.1.2 SPH discretization

In SPH, a continuous fluid domain is discretized into a set of particles, each carrying physical properties such as mass, density, velocity, and pressure. The fundamental principle of SPH, illustrated in Fig. 1 lies in the interpolation of field variables using a smoothing kernel function, which enables the approximation of continuous functions and their derivatives through a weighted summation over neighboring particles.

In the discretized SPH framework, the continuous integral is approximated as a summation over neighboring particles. For a particle i , the value of the function $f_i = f(\mathbf{r}_i)$ is computed as:

$$f_i = \sum_j f_j W_{ij} V_j = \sum_j f_j W_{ij} \frac{m_j}{\rho_j}, \quad (3)$$

where j denotes neighboring particles within the support domain of particle i , $W_{ij} = W(|\mathbf{r}_{ij}|, h)$ is the smooth kernel function evaluated at the distance $|\mathbf{r}_{ij}| = |\mathbf{r}_i - \mathbf{r}_j|$. Here, m_j , ρ_j , and V_j represent the mass, density, and volume of particle j , respectively.

The choice of kernel function is critical in SPH, as it governs the accuracy, stability, and computational efficiency of the method. In this study, the Wendland C2 kernel is adopted due to its superior stability and smoothness properties. The kernel is defined as:

$$W(q) = \begin{cases} C(1-q)^4(1+4q), & \text{if } 0 \leq q < 1 \\ 0, & \text{if } q \geq 1 \end{cases} \quad (4)$$

Here, $q = |\mathbf{r}_{ij}|/h$, $C = 21/(2\pi h^3)$ in 3D, and the smoothing length $h = 3.015\Delta_{SPH}$, where Δ_{SPH} is the initial particle spacing. The discretization of gradient, divergence, and Laplacian operators in SPH can be derived using various formulations and the derivations are well presented in some literatures.

2.1.3 EISPH for fluid flow

In this study, the explicit incompressible SPH (EISPH) method is employed to solve the governing equations of fluid flow. The Incompressible SPH method, which is generally based on the first-order projection approach, ensures mass and momentum conservation through a three-step process.

2.1.3.1 Prediction step (Intermediate velocity calculation) The intermediate velocity field ($\mathbf{u}^*$) is computed by solving the momentum equation while neglecting the

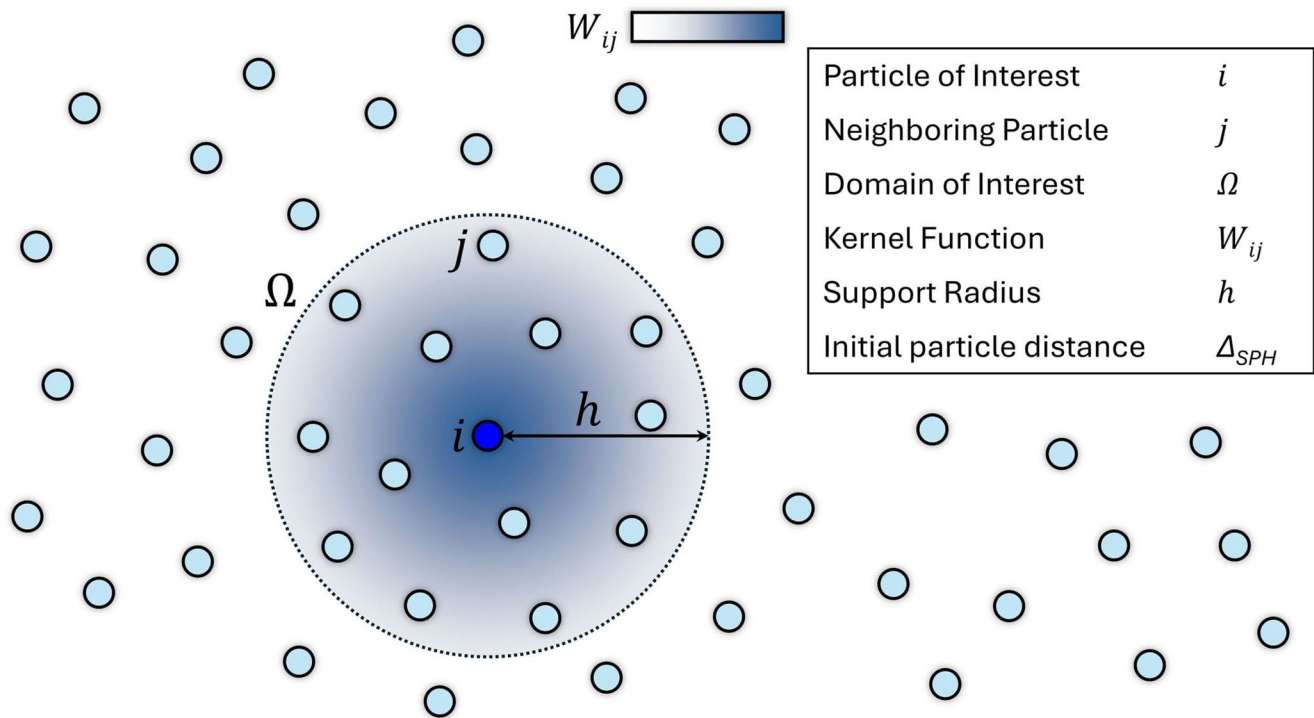


Fig. 1 Schematic Representation of SPH Discretization

pressure contribution. This step accounts for advection and external forces:

$$\frac{\mathbf{u}_i^* - \mathbf{u}_i^n}{\Delta t} = \mathbf{g} + \mu_i \sum_j \Pi_{ij} \nabla_i W_{ij} \frac{m_j}{\rho_j}, \quad (5)$$

where the operator Π_{ij} is represented as [41]:

$$\Pi_{ij} = 2(dim + 2) \frac{\mathbf{u}_{ij} \cdot \mathbf{r}_{ij}}{|\mathbf{r}_{ij}|^2 + \eta^2}, \quad (6)$$

where the dim denotes the spatial dimension and η is a small constant set to be $0.1h$ to avoid singularities.

2.1.3.2 Pressure Poisson equation (PPE) The pressure field P_i^{n+1} is determined by enforcing the incompressibility condition, ensuring that the divergence of the velocity field remains zero. The PPE is solved as:

$$\begin{aligned} \sum_{j=1}^N (P_i^{n+1} - P_j^{n+1}) \frac{2\mathbf{r}_{ij} \cdot \nabla W(\mathbf{r}_{ij}, h)}{|\mathbf{r}_{ij}|^2 + \eta^2} \frac{m_j}{\rho_j} \\ = \frac{\sum_j \rho_i (\mathbf{u}_j^* - \mathbf{u}_i^*) \cdot \nabla W(\mathbf{r}_{ij}, h) \frac{m_j}{\rho_j}}{\Delta t}, \end{aligned} \quad (7)$$

The PPE can be solved in matrix form but, recent studies have shown that an explicit formulation can also accurately capture fluid dynamics, heat transfer, and fluid–structure interaction (FSI) phenomena when the time step is suitably constrained. Adopting this approach, the explicit form of the PPE is written as:

$$\begin{aligned} \sum_{j=1}^N (P_i^{n+1} - P_j^n) \frac{2\mathbf{r}_{ij} \cdot \nabla W(\mathbf{r}_{ij}, h)}{|\mathbf{r}_{ij}|^2 + \eta^2} \frac{m_j}{\rho_j} = \\ \frac{\sum_j \rho_i (\mathbf{u}_j^* - \mathbf{u}_i^*) \cdot \nabla W(\mathbf{r}_{ij}, h) \frac{m_j}{\rho_j}}{\Delta t}. \end{aligned} \quad (8)$$

2.1.3.3 Correction step (Velocity update) The final velocity $\mathbf{u}_i^{n+1}$ is obtained by correcting the intermediate velocity with the computed pressure gradient as:

$$\frac{\mathbf{u}_i^{n+1} - \mathbf{u}_i^*}{\Delta t} = - \sum_j 2 \frac{P_i \rho_i + P_j \rho_j}{\rho_i + \rho_j} \nabla_i W_{ij} \frac{m_j}{\rho_i \rho_j}. \quad (9)$$

In this study, kernel gradient correction (KGC) method has been used to compute pressure force, and the corrected formulation becomes:

$$\frac{\mathbf{u}_i^{n+1} - \mathbf{u}_i^*}{\Delta t} = - \sum_j 2 \frac{P_i \rho_i + P_j \rho_j}{\rho_i + \rho_j} L_i \cdot W_{ij} \frac{m_j}{\rho_i \rho_j}, \quad (10)$$

where the matrix $\mathbf{L}_i$ is commonly referred to as the correction matrix or renormalization matrix in the literature, and is defined as:

$$L_i := \left[\sum_j \mathbf{r}_{ji} \otimes \nabla_i W_{ij} \frac{m_j}{\rho_j} \right]^{-1}. \quad (11)$$

2.1.4 SPH boundary condition

2.1.4.1 Free surface detection To impose free surface boundary conditions, the identification of free surface particles is required. Various methods have been proposed for defining free surface particles, and in this study free surface particles are identified by the criterion based on the minimum eigenvalue (λ_i) of the inverse matrix of renormalization matrix ($\mathbf{L}_i^{-1}$) [42].

The classification is as follows:

- Free surface particles : $\lambda_i < 0.6$.
- Interface particles : $0.6 \leq \lambda_i < 0.95$
- Interior particles : $\lambda_i \geq 0.95$.

2.1.4.2 Free surface boundary condition Several strategies have been developed to impose Dirichlet boundary conditions on the free surface in ISPH [43–45]. However, explicitly enforcing the Dirichlet condition (e.g., zero pressure) can lead to instabilities near the free surface, as the incompressibility condition inherently satisfies the boundary implicitly [44].

In the present study, we adopt an implicit enforcement strategy for the free surface boundary when solving the pressure Poisson equation (PPE) [43]. In the free surface particles, particles follow the following equation:

$$\sum_{j=1}^N (P_i^{n+1} - P_O) \frac{2\mathbf{r}_{ij} \cdot \nabla W(\mathbf{r}_{ij}, h) m_j}{|\mathbf{r}_{ij}|^2 + \eta^2} \frac{m_j}{\rho_j} = \frac{\sum_j \rho_j \mathbf{u}_{ji}^* \cdot \nabla W(\mathbf{r}_{ij}, h) \frac{m_j}{\rho_j}}{\Delta t}, \quad (12)$$

where, the P_O represent the ambient pressure corresponding to the free-surface particles. In this study, the P_O is set as zero for free-surface particles.

2.1.4.3 Solid wall boundary condition To model the wall boundary, fixed ghost particles with the same material prop-

erties as the fluid are employed. The physical pressure field is computed using a generalized boundary condition [46]:

$$P_{i,w} := \frac{\sum_{j \in \Omega_f} P_j W_{ij} + \mathbf{g} \cdot \sum_{j \in \Omega_f} \rho_j \mathbf{r}_{ij} W_{ij}}{\sum_{j \in \Omega_f} W_{ij} + \eta^2}. \quad (13)$$

Here, the subscripts f and w indicates fluid and wall particles. In this study, slip boundary condition is mostly used for the wall boundary. The slip boundary condition for velocity is enforced by neglecting the contribution of the viscous term [46].

2.1.5 Quasi-Lagrangian EISPH scheme

In this study, the governing equations are reformulated using a quasi-Lagrangian approach, based on the demonstrated enhanced numerical stability and accuracy through similar strategies [10, 47]. This reformulation employs the Arbitrary Lagrangian-Eulerian (ALE) time derivative, defined as:

$$\frac{df}{dt} := \frac{\partial f}{\partial t} + (\mathbf{u} + \delta \mathbf{u}) \cdot \nabla f = \frac{Df}{Dt} + \delta \mathbf{u} \cdot \nabla f. \quad (14)$$

Here, Dt/D denotes the Lagrangian derivative, while $\delta \mathbf{u}$ represents the particle shifting velocity. In the pure Lagrangian framework, $\delta \mathbf{u} = 0$. The particle shifting velocity is introduced to improve particle distribution and reduce tensile instability, and it is calculated using a modified concentration gradient as described by Fick's law [48] with the proportional constant D'_i :

$$\nabla_i C_i = \sum_j \left[1 + 0.2 \left(\frac{W_{ij}}{W_{ij}(\Delta x)} \right)^4 \right] \frac{m_j}{\rho_j} \nabla_i W_{ij}, \quad (15)$$

$$\delta \mathbf{u}_i = -D'_i \nabla_i C_i \quad (16)$$

For interior particles, the shifting velocity is further refined using the following expression:

$$\delta \mathbf{u}_i = \delta \mathbf{u}_{i,\text{interior}} := -\kappa h_i |\mathbf{u}_i|_{\max} \nabla_i C_i, \quad (17)$$

To preserve kinematic boundary conditions and prevent unphysical flow distortion near free surfaces, no shifting is applied to particles identified as free surface particles. For the interface particles, between the free surface particles and interior particles, an optimized shifting strategy is applied that restricts displacement to the tangential direction by eliminating the normal component [49].

$$\delta \mathbf{u}_i = \delta \mathbf{u}_{i,\text{interface}} := -\kappa h_i |\mathbf{u}_i|_{\max} (\mathbf{I} - \mathbf{n}_i \otimes \mathbf{n}_i) \cdot \nabla_i C_i. \quad (18)$$

Here, $\mathbf{n}_i$ is the normal vector of the particle which can be derived as:

$$\mathbf{n}_i = - \frac{\sum_j (\lambda_j - \lambda_i) \mathcal{L}_i \cdot \nabla_i W_{ij} \frac{m_j}{\rho_j}}{\left| \sum_j (\lambda_j - \lambda_i) \mathcal{L}_i \cdot \nabla_i W_{ij} \frac{m_j}{\rho_j} \right|} \quad (19)$$

To ensure that particle shifting remains within reasonable bounds, the magnitude of $\delta \mathbf{u}_i$, is limited to half of the particle's physical velocity:

$$\delta \mathbf{u}_i = \begin{cases} \delta \mathbf{u}_i & \text{if } |\delta \mathbf{u}_i| < 0.5 |\mathbf{u}_i|, \\ -0.5 |\mathbf{u}_i| \frac{\delta \mathbf{u}_i}{|\delta \mathbf{u}_i|} & \text{otherwise,} \end{cases} \quad (20)$$

Within the ALE framework, and considering the shifting velocity, it is necessary to account for the deviation from purely Lagrangian motion by including additional convection terms ($\delta \mathbf{u}_i \bullet \nabla_i \mathbf{u}_i$). To further stabilize the numerical solution and improve accuracy, these convective terms are discretized using an upwind interpolation scheme [50, 51]. The upwind interpolation employs an anisotropic weighting function defined as [50]:

$$W_{ij}^u = \begin{cases} W_{ij} & \phi_{ij} \geq \pi/2 \\ \epsilon W_{ij} & \phi_{ij} < \pi/2 \end{cases} \quad (21)$$

where $\epsilon = 0.01$ is a small constant and, ϕ_{ij} is the angle between the relative position vector $\mathbf{r}_{ij}$ and the particle shifting direction:

$$\phi_{ij} = \arccos \left(\frac{\mathbf{r}_{ij}}{\|\mathbf{r}_{ij}\|} \cdot \frac{\delta \mathbf{u}_i}{\|\delta \mathbf{u}_i\|} \right) \quad (22)$$

2.2 Governing equation for structure

2.2.1 Governing equation in classical continuous mechanics

The governing equation for solid mechanics is formulated based on the principle of momentum conservation within a continuous medium. It is expressed as [26]:

$$\rho(\mathbf{x}) \dot{\mathbf{u}}(\mathbf{x}, t) = \mathbf{L}(\mathbf{x}, t) + \mathbf{b}(\mathbf{x}, t), \quad (23)$$

where $\mathbf{u}(\mathbf{x}, t)$ denotes the displacement field, $\mathbf{L}(\mathbf{x}, t)$ is the internal force term, and $\mathbf{b}(\mathbf{x}, t)$ accounts for the body force per unit volume. The position vector $\mathbf{x} = (x, y, z)$ represents the location of arbitrary point with respect to the reference configuration at time t .

In classical continuous mechanics, internal force is defined as:

$$\mathbf{L}(\mathbf{x}, t) := \nabla \cdot \boldsymbol{\sigma}(\mathbf{x}, t) \quad (24)$$

where $\boldsymbol{\sigma}$ represents the Cauchy stress tensor.

2.2.2 Peridynamics (PD) discretization

Peridynamics (PD) is a reformulation of classical continuum mechanics that replaces spatial derivatives with integral equations. This allows for the natural modeling of discontinuities, such as cracks and fractures, which are challenging to handle using classical approaches [23, 26].

Unlike traditional continuum mechanics, PD models internal forces as integral interactions over a finite horizon δ instead of relying on differential operators as in the Fig. 2. The most general form of the peridynamic equation of motion is given by the state-based formulation:

$$\rho(\mathbf{x}) \dot{\mathbf{u}}(\mathbf{x}, t) = \mathbf{L}^{PD}(\mathbf{x}, t) + \mathbf{b}(\mathbf{x}, t), \quad (25)$$

where the peridynamic internal force density is defined as [27]:

$$\mathbf{L}^{PD}(\mathbf{x}, t) := \int_{\mathcal{H}_x} (\mathbf{t}(\mathbf{u}' - \mathbf{u}, \mathbf{x}' - \mathbf{x}, t) - \mathbf{t}'(\mathbf{u} - \mathbf{u}', \mathbf{x} - \mathbf{x}', t)) dV_{\mathbf{x}'} \quad (26)$$

Here, $\mathcal{H}_x$ indicates the integral domain around point $\mathbf{x}$, represented by the horizon δ and, $\mathbf{t}$ is the force density function. To numerically solve the peridynamic equation of motion, the integral formulation is discretized using a set of material points. The integral in the governing equation is replaced with a summation over neighboring discrete material points [52, 53].

$$\mathbf{L}^{PD}(\mathbf{x}_A, t) = \sum_{B \in H_A} (\mathbf{t}_{AB}(\mathbf{u}_B - \mathbf{u}_A, \mathbf{x}_B - \mathbf{x}_A, t) - \mathbf{t}_{BA}(\mathbf{u}_A - \mathbf{u}_B, \mathbf{x}_A - \mathbf{x}_B, t)) V_B \quad (27)$$

$\mathbf{x}_A = (x_A, y_A, z_A)$ and $\mathbf{x}_B = (x_B, y_B, z_B)$ denote the position vectors of material points A and B , respectively. In this study, horizon $\delta = 3.015 \Delta_{PD}$, where Δ_{PD} is the initial particle spacing.

2.2.3 Ordinary state-based PD for structure analysis

In this study, the ordinary state-based peridynamics (OSPD) is employed, where the force states $\mathbf{t}$ and $\mathbf{t}'$ are defined to act in opposite directions with unequal magnitudes. This formulation inherently satisfies the angular momentum conservation principle without requiring additional constraints.

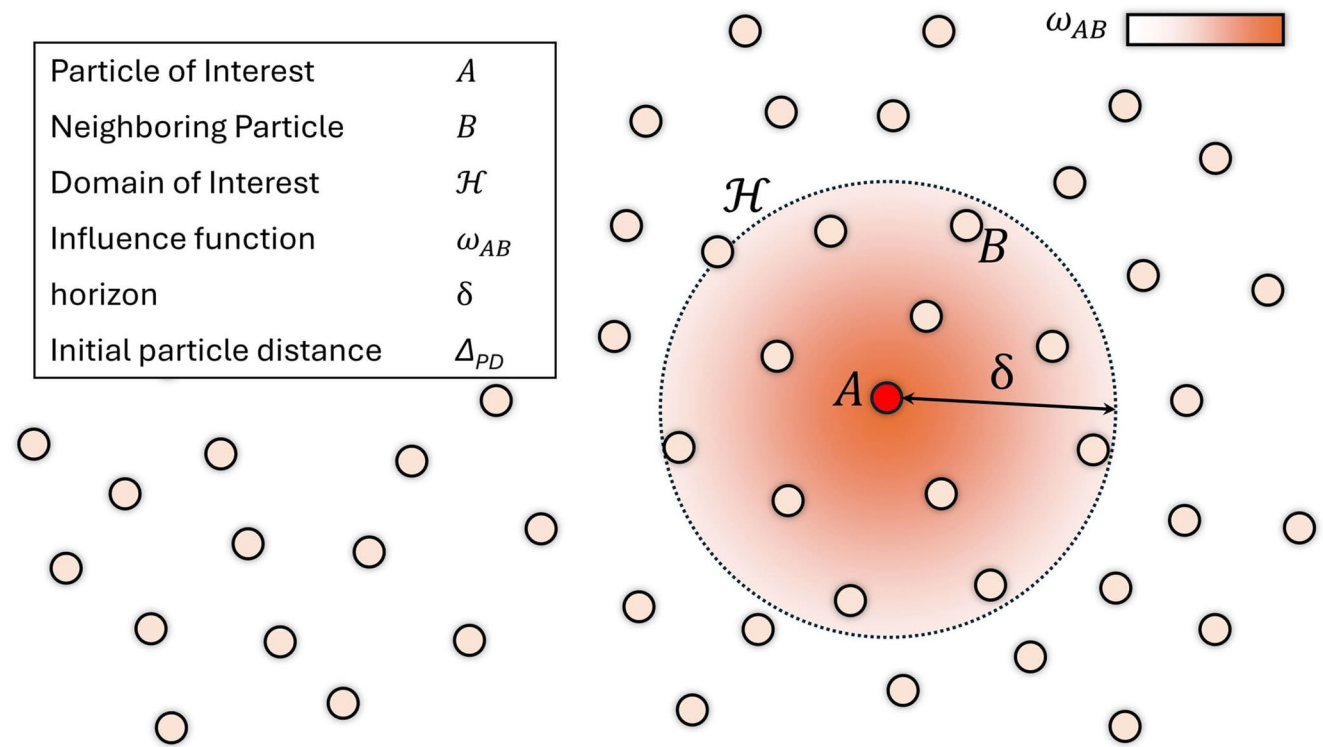


Fig. 2 Schematic Representation of PD Discretization

In OSPD, the force density vector at material point $\mathbf{x}_A$ is analytically linked to the strain energy density (SED) function [54]:

$$\mathbf{t}_{AB}(\mathbf{u}_B - \mathbf{u}_A, \mathbf{x}_B - \mathbf{x}_A, t) = \frac{1}{V_B} \frac{\partial W_A}{\partial (|\mathbf{y}_B - \mathbf{y}_A|)} \frac{\mathbf{y}_B - \mathbf{y}_A}{|\mathbf{y}_B - \mathbf{y}_A|}, \quad (28)$$

where, the $\mathbf{y}^? \mathbf{x} + \mathbf{u}(\mathbf{x}, t)$ represents the deformed position of the material point and W_A is the strain energy density (SED) function at the material point A .

For an elastic isotropic material, the strain energy density function is decomposed into dilatational and distortional components [52, 55]:

$$W_A = W_A^K + W_A^\mu, \quad (29)$$

where W_A^K represents the dilatational (volumetric) deformation energy and W_A^μ represents the distortional (shear) deformation energy. The dilatational SED is given by:

$$W_A^K = a_K \theta_A^2, \quad (30)$$

where θ_A denotes the dilatation at material point A , which is discretized as:

$$\theta_A = d \sum_{B \in H_A} \omega_{AB} (|\mathbf{y}_B - \mathbf{y}_A| - |\mathbf{x}_B - \mathbf{x}_A|) V_B, \quad (31)$$

where d is a parameter, ω_{AB} is the influence function. The distortional SED accounts for shear deformation and is expressed as:

$$W_A^\mu = \left\{ b \sum_{B \in H_A} \omega_{AB} (|\mathbf{y}_B - \mathbf{y}_A| - |\mathbf{x}_B - \mathbf{x}_A|)^2 V_B - a_\mu \theta_A^2 \right\}. \quad (32)$$

By differentiating the SED, the force density vector at material point A can be rewritten as:

$$\mathbf{t}_{AB} = 2d\omega_{AB} (a\theta_A) + 2b\omega_{AB} (|\mathbf{y}_B - \mathbf{y}_A| - |\mathbf{x}_B - \mathbf{x}_A|). \quad (33)$$

This formulation explicitly accounts for both volumetric and shear deformation effects in the PD interaction, where $a^? a_K - a_\mu$. A commonly used influence function in PD literature includes both power-law and Gaussian forms, given as:

$$\omega_{AB} = R^n \text{ (power-law form)} \quad (34)$$

$$\omega_{AB} = e^{-R^2} \text{ (Gaussian form)} \quad (35)$$

where, $R := \delta / |\mathbf{x}_B - \mathbf{x}_A|$ represents the normalized interaction distance, n is an exponent, commonly chosen to be 1 in most applications.

To ensure consistency with classical continuum mechanics, the coefficients a , b , and d can be derived by equating

the peridynamic expressions for dilatation and strain energy density (SED) with those of classical linear elasticity [54]. The resulting expressions for these coefficients at 3-dimensions are:

For **power-law form** influence function :

$$a = \frac{K}{2} - \frac{\pi b \delta^5}{9}, \quad b = \frac{15\mu}{8\pi\delta^5} (5+n), \quad d = \frac{3}{4\pi\delta^4} (n+4), \quad (36)$$

For **Gaussian form** influence function :

$$a = \frac{K}{2} - \frac{\pi b \delta^5}{9}, \quad b = \frac{15\mu}{8\pi\delta^5} \left(\frac{8e}{3e\sqrt{\pi}\operatorname{erf}(1) - 10} \right), \quad (37)$$

$$d = \frac{3}{4\pi\delta^4} \left(\frac{2e}{e-2} \right)$$

In this study, different forms of influence functions are considered:

- **Inverse function:** $\omega_{AB} = 1/R$ (hereafter denoted as ω_{-1}),
- **Constant function:** $\omega_{AB} = 1$ (hereafter denoted as ω_0),
- **Linear function:** $\omega_{AB} = R$ (hereafter denoted as ω_1),
- **Gaussian function:** $\omega_{AB} = e^{-R^2}$ (hereafter denoted as ω_g).

2.2.4 PD correction scheme

To improve numerical accuracy in peridynamic simulations, especially near boundaries or irregular particle distributions, correction schemes are essential. Two key strategies employed in this study are the Volume Correction Scheme (VCS) and the Surface Correction Scheme (SCS).

2.2.4.1 Volume correction scheme (VCS) The precise calculation of volume is important to ensure the numerical calculation accuracy of the PD models. Various volume/area correction schemes are proposed [56], and in this study, the most general and simple form is utilized [57].

$$V_B^C = \begin{cases} V_B & |\mathbf{x}_{AB}| \leq \delta - 0.5\Delta_{PD} \\ \frac{\delta - (|\mathbf{x}_{AB}| - 0.5\Delta_{PD})}{\Delta_{PD}} V_B & \delta - 0.5\Delta_{PD} < |\mathbf{x}_{AB}| \leq \delta + 0.5\Delta_{PD} \\ 0 & else \end{cases} \quad (38)$$

2.2.4.2 Surface correction scheme (SCS) The peridynamic material parameters a_κ , a_μ , b , and d , which govern the force-stretch relations, are determined by evaluating the dilatation and strain energy density of a material point whose horizon is entirely within the material. While a_κ and a_μ remains unchanged, b and d are influenced by the integration domain defined by the horizon. As a result, adjustments

to b and d are necessary when the material point is near a free surface or material interface to account for the altered interaction domain.

To account for deviations in dilatation and strain energy density (SED) near free surface or material interfaces, correction factors are introduced based on theoretical and numerical evaluations under prescribed deformation modes. These factors serve to restore the accuracy of the non-local integral terms that are influenced by the truncated interaction domain. Specifically, dilatation correction factors can be obtained by applying a prescribed isotropic expansion, while SED correction factors are derived under a pure shear deformation field. For a more comprehensive explanation of the derivation, the readers are referred to [54].

The correction factor vectors for dilatation and SED can be written as:

$$\mathbf{g}_{d,A} = [D_{x,A} \ D_{y,A} \ D_{z,A}]^T, \quad \mathbf{g}_{d,A} = [S_{x,A} \ S_{y,A} \ S_{z,A}]^T \quad (39)$$

However, since peridynamics is inherently formulated based on pairwise interactions, these correction factors must be extended to reflect bond-based contributions:

$$g_{\beta,A,B} = [\bar{g}_{x,\beta,A,B}, \bar{g}_{y,\beta,A,B}, \bar{g}_{z,\beta,A,B}]^T = \frac{g_{\beta,A} + g_{\beta,B}}{2} \quad (40)$$

with $\beta = d, b$. The scalar pairwise correction factor is obtained by projecting the correction vector onto the unit relative position vector to ensure the correction is consistently applied along the deformation direction:

$$G_{\beta,A,B} = \left([n_x/\bar{g}_{x,\beta,A,B}]^2 + [n_y/\bar{g}_{y,\beta,A,B}]^2 + [n_z/\bar{g}_{z,\beta,A,B}]^2 - 1/2 \right) \quad (41)$$

After accounting for the surface effects, the corrected expressions for the dilatation and the strain energy density can be given by:

$$\theta_A = d \sum_{B \in H_A} G_{d,A,B} \omega_{AB} (|\mathbf{y}_B - \mathbf{y}_A| - |\mathbf{x}_B - \mathbf{x}_A|) V_B \quad (42)$$

$$W_A^\mu = \left\{ b \sum_{B \in H_A} G_{b,A,B} \omega_{AB} (|\mathbf{y}_B - \mathbf{y}_A| - |\mathbf{x}_B - \mathbf{x}_A|)^2 V_B - a_\mu \theta_A^2 \right\} \quad (43)$$

2.2.5 PD fracture model

In PD formulation, the stretch is defined as the relative elongation of the bond between two interacting material points [58].

$$s_{AB} := \frac{|\mathbf{y}_B - \mathbf{y}_A| - |\mathbf{x}_B - \mathbf{x}_A|}{|\mathbf{x}_B - \mathbf{x}_A|} \quad (44)$$

The stretch s_{AB} captures the local strain within the PD interaction. In this paper, material failure is governed in a simple and general manner using an irreversible damage criterion based on a critical stretch threshold. The interaction between material points A and B is permanently removed, when the stretch s_{AB} exceeds the critical stretch [58, 59], with the experimentally predictable energy release G_0 and bulk modulus K .

$$s_{AB} \geq s_C = \sqrt{5G_0/9K\delta} \Rightarrow \omega_{AB} = 0 \quad (44)$$

3 Coupled SPH-PD numerical framework

In this study, the PD and SPH models are monolithically coupled within a unified GPU framework. Each model is explicitly implemented based on single GPU parallelization according to their respective time step constraints. The two solvers are coupled in a serial manner, ensuring accurate and efficient interaction between the two methods. This section presents the detailed numerical boundary conditions, and the computational coupling strategy adopted in the framework, with the aim of maintaining both numerical accuracy and computational efficiency in simulating complex FSI problems.

3.1 SPH-PD numerical boundary condition

In the fluid structure interaction problem, the SPH particles representing fluid domain and PD particles modeling structural domain are faced together. At the fluid-structure interface, these particles interaction to ensure consistent momentum exchange and accurate force transmission across the interface as in Fig. 3. The coupling is implemented through pressure-based interaction.

Firstly, the pressure field must be applied to the PD particles to accurately compute correct pressure force term. In this study, the pressure field is extrapolated from the surrounding SPH particles, considering the acceleration of the PD particle, and is expressed as follows [46]:

$$P_{A,PD} := \frac{\sum_{j \in \Omega_{f,SPH}} P_j W_{Aj} + (\mathbf{g} - \mathbf{a}_A) \cdot \sum_{j \in \Omega_{f,SPH}} \rho_j \mathbf{r}_{Aj} W_{Aj}}{\sum_{j \in \Omega_{f,SPH}} W_{Aj} + \eta^2}, \quad (45)$$

with the acceleration of PD particle $\mathbf{a}_A$.

- **PD particle $\rightarrow$ SPH particle.**
- For an SPH particle with neighboring PD particles, it receives pressure forces from both SPH and PD neighbors,

enabling the fluid to respond to both fluid and structural surroundings. The pressure interaction between SPH particles is computed as described in Sect. 2. In addition to this, the pressure force contribution from PD particles to SPH particles is calculated as follows and added to the pressure force term:

$$\mathbf{f}_{PD \rightarrow SPH} = - \sum_{B \in \Omega_{PD}} 2 \frac{P_i \rho_i + P_B \rho_B}{\rho_i + \rho_B} \nabla_i W_{iB} \frac{m_B}{\rho_i \rho_B} \quad (46)$$

- **SPH particle $\rightarrow$ PD particle.**
- For a PD particle, it receives pressure forces only from neighboring SPH particles. Internal interactions within the structural domain are governed solely by the PD formulation, while the fluid-induced pressure loading is introduced externally through the coupling with SPH particles:

$$\mathbf{f}_{SPH \rightarrow PD} = - \sum_{j \in \mathcal{H}_{A,SPH}} 2 \frac{P_A \rho_A + P_j \rho_j}{\rho_A + \rho_j} \nabla_A W_{Aj} \frac{m_j}{\rho_A \rho_j} \quad (47)$$

To compute pressure from SPH particles to PD particles, the kernel function between SPH and PD particles and pressure of PD particles are computed samely with SPH formulation as Eq. 4. This formulation ensures two-way coupling between the fluid and structure, while preserving the distinct physical laws governing each domain.

3.2 SPH-PD computational coupling strategy

In this study, the SPH and PD solvers are coupled in a parallel and explicit manner to efficiently simulate the dynamics of fluid–structure interaction. The standalone computational procedures for each solver are illustrated in Fig. 4. When executed independently, each solver follows its own algorithm without the inclusion of the coupling step highlighted in the blue boxes. When the two solvers are coupled, the blue-boxed operations are activated to facilitate fluid–structure data exchange and interaction force computation.

For the fluid domain, the SPH method is formulated using a first-order projection scheme for time integration [60, 61]. After initialization, nearest neighbor particle search (NNPS) is performed at each time step to identify interacting particles. Subsequently, the governing equations for mass and momentum conservation are solved via the projection approach, which separates the computation of intermediate velocities and following by the pressure force correction.

For the PD method, both the first-order Euler method and the second-order predictor-corrector scheme are considered. However, as elaborately demonstrated in the numerical analyses in Sect. 4, the use of the Euler scheme led to noticeable checkerboard-like oscillations in the displacement and

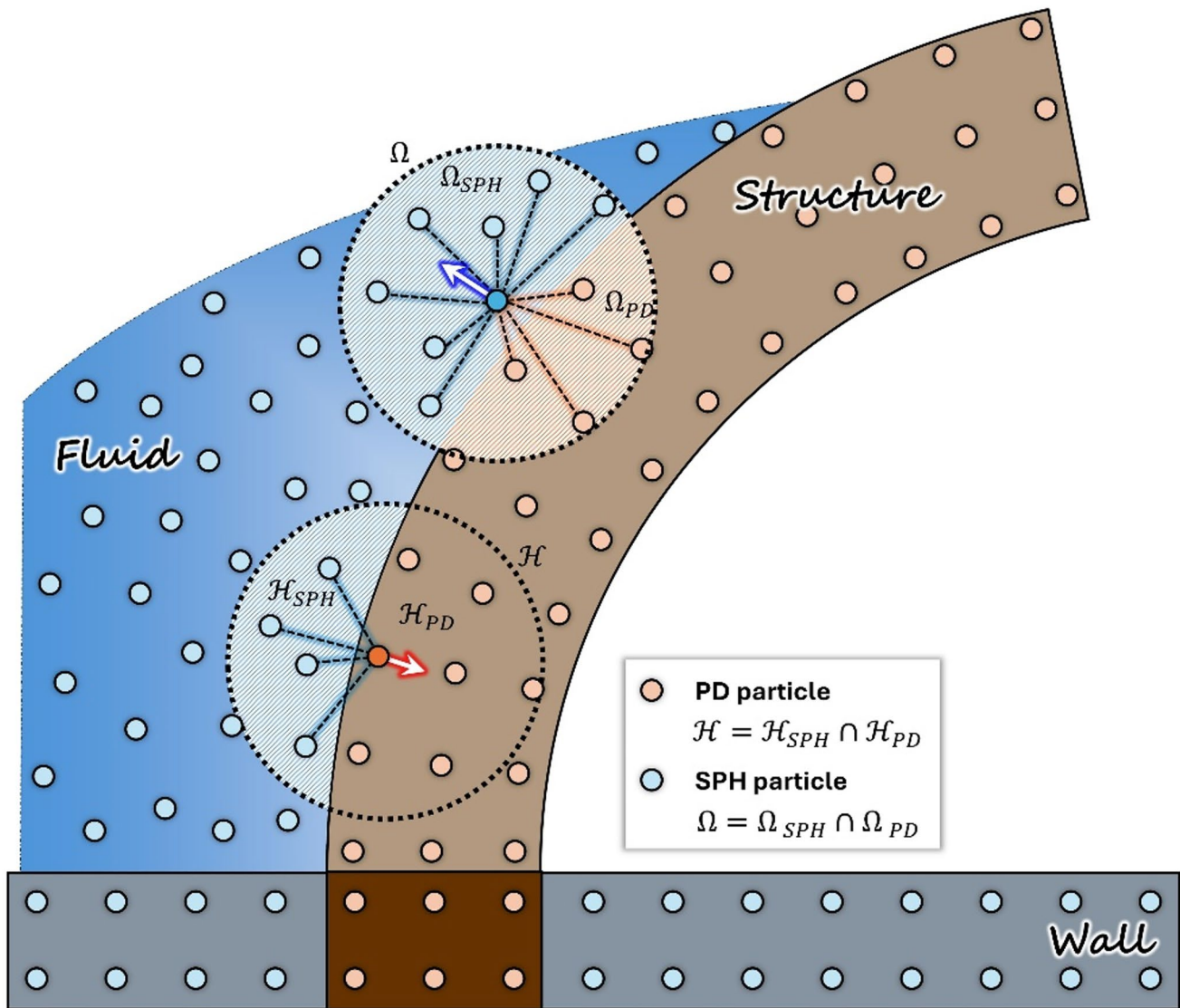


Fig. 3 Conceptual Schematic of Fluid–Structure Coupling via SPH and PD

dilatation fields. To mitigate these issues, a higher-order scheme was adopted, and the predictor–corrector method was selected for all simulations in this study.

In contrast to SPH, where particle interactions evolve significantly over time, structural deformation in the PD domain remains relatively smooth and continuous, particularly in the elastic regime. As a result, the nearest neighbor particle search (NNPS), which is computationally expensive, is performed only once at the initial time step. Similarly, the surface correction term which is introduced in Sect. 2 to account for truncated bonds near boundaries is also computed once at initialization. After these preprocessing steps, the dilatation, and PD forces are evaluated sequentially at each time step.

Each method operates under its own characteristic time step, which is either predefined or determined based on time step constraints specific to the SPH and PD formulations:

$$\Delta t_f = \min_i \left(C_{CFL,f} \frac{h_i}{\max |\mathbf{u}_i|}, 0.125 \frac{h_i^2 \rho_0}{\mu_0} \right), \quad (48)$$

$$\Delta t_s = \min_i \left(C_{CFL,s} \frac{h_i}{c^s} \right). \quad (49)$$

Here ρ_0 and μ_0 represent the reference density and dynamic viscosity of the fluid, respectively. c^s refers to the speed of sound of structure ($c^s = \sqrt{K^s / \rho^s}$), where K^s and ρ^s denote the stiffness and density of the solid,

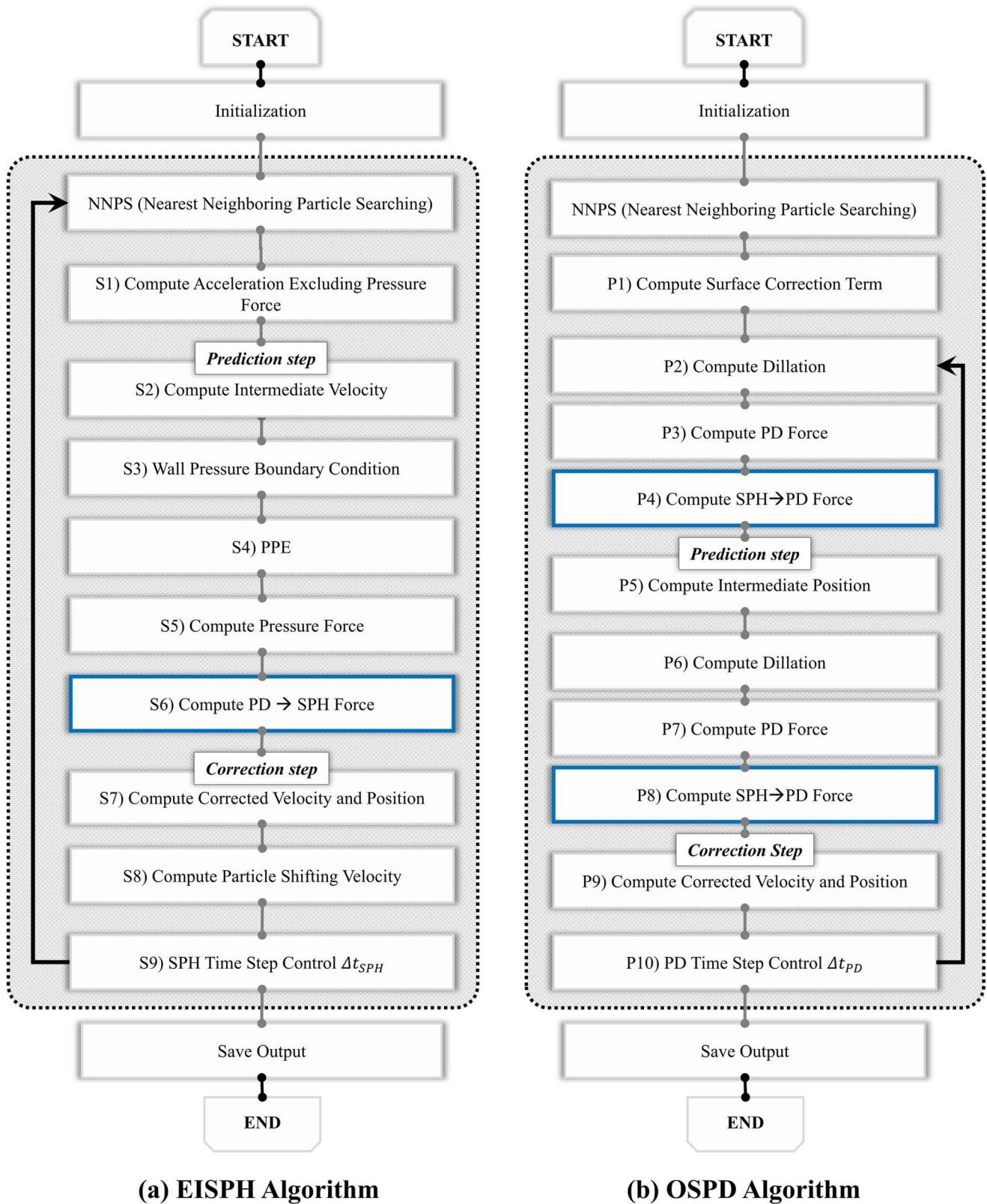


Fig. 4 Stand Alone Algorithms of EISPH and OSPD

respectively. $C_{CFL,f}$ and $C_{CFL,s}$ symbolize the CFL number for fluid and structure, respectively.

In conventional SPH and PD coupling frameworks, the global time step is usually determined by the most restrictive condition, that is, the minimum of the SPH and PD time steps, to maintain numerical stability [21, 36, 62]. Although this approach ensures stable integration between the fluid and solid solvers, it often results in unnecessary computational inefficiency, especially when the two solvers have very different stability limits. In particular, the solver with a larger allowable time step becomes needlessly restricted, as noted in recent studies [39, 63].

To improve computational efficiency while preserving numerical stability, this study adopts a subcycling-like time integration scheme. Unlike conventional approaches that rely on a unified minimum time step, the proposed method allows each solver to advance using its own time step.

Specifically, the PD solver, which generally requires a smaller time step due to the stiffness of solid materials, progresses through a series of updates until its accumulated time slightly exceeds that of the SPH solver. At that point, both solvers advance together: the PD solver proceeds using the excess time, while the SPH solver advances by one full time step. This approach ensures temporal synchronization between the two solvers. The time integration scheme can be summarized as follows:

- The PD model advances a loop using its own time step.
- When the accumulated PD time exceeds the SPH time step, both the PD and SPH solvers are advanced together using the remaining surplus time for PD and one full step for SPH.
- After synchronization, the cycle repeats from the updated simulation time.

This method enables the PD solver to resolve high-frequency structural dynamics with accuracy, while avoiding the computational penalty that would occur if the SPH solver were forced to operate under the smaller PD time step. As a result, the proposed approach ensures both computational efficiency and stable simulation of fluid structure interaction.

The overall structure of the coupled algorithm is shown in Fig. 6. As depicted, after initialization, the nearest NNPS is performed at each SPH computing step. Although the PD solver alone would require NNPS only once due to relatively stable neighbor configurations in elastic problems, in FSI simulations it is necessary to identify interactions between SPH and PD particles. Therefore, NNPS is conducted across both SPH and PD particles at every SPH time step. Following neighbor search, the SPH and PD domains are updated using their respective solvers, as outlined in the flowchart. These operations are executed sequentially within the time-marching loop shown in Fig. 5.

In this study, a CPU and GPU heterogeneous computing strategy is adopted. The initialization and output operations are handled on the CPU, while all numerical computations are carried out on the GPU. As a result, memory transfer from CPU to GPU occurs only during initialization, and GPU to CPU transfer is performed only at the designated output intervals. This reduces memory transfer overhead and improves overall performance. The GPU implementation in this study follows a similar strategy to that used in our previous works [64, 65]. All simulations in this study were conducted on a workstation featuring an NVIDIA Tesla V100-PCIE-32GB GPU and an Intel(R) Xeon(R) Silver 4210R CPU @ 2.40 GHz.

4 Numerical validation

4.1 Validation of EISPH model

To accurately model SPH–PD for fluid–structure interaction, it is essential to ensure that the pressure load exerted on the structure is properly validated within the EISPH framework. To achieve this, the three-dimensional dam-break problem is employed as a benchmark test to assess the convergence and accuracy of the time evolution of the pressure load acting on a rigid structure.

As illustrated in Fig. 7, we consider the dam break problem with a small obstruction which is one of the well-established benchmark in the SPH community [66–68]. The densities of the water, and obstruction are all set to $\rho_0 = 1000\text{kg/m}^3$ and $\rho^s = 1000\text{kg/m}^3$, while the

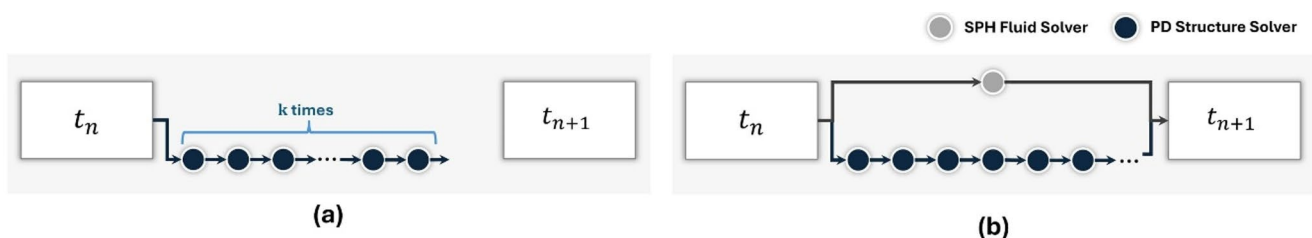


Fig. 5 Temporal Integration Strategy for EISPH–OSPD Coupling with Subcycling

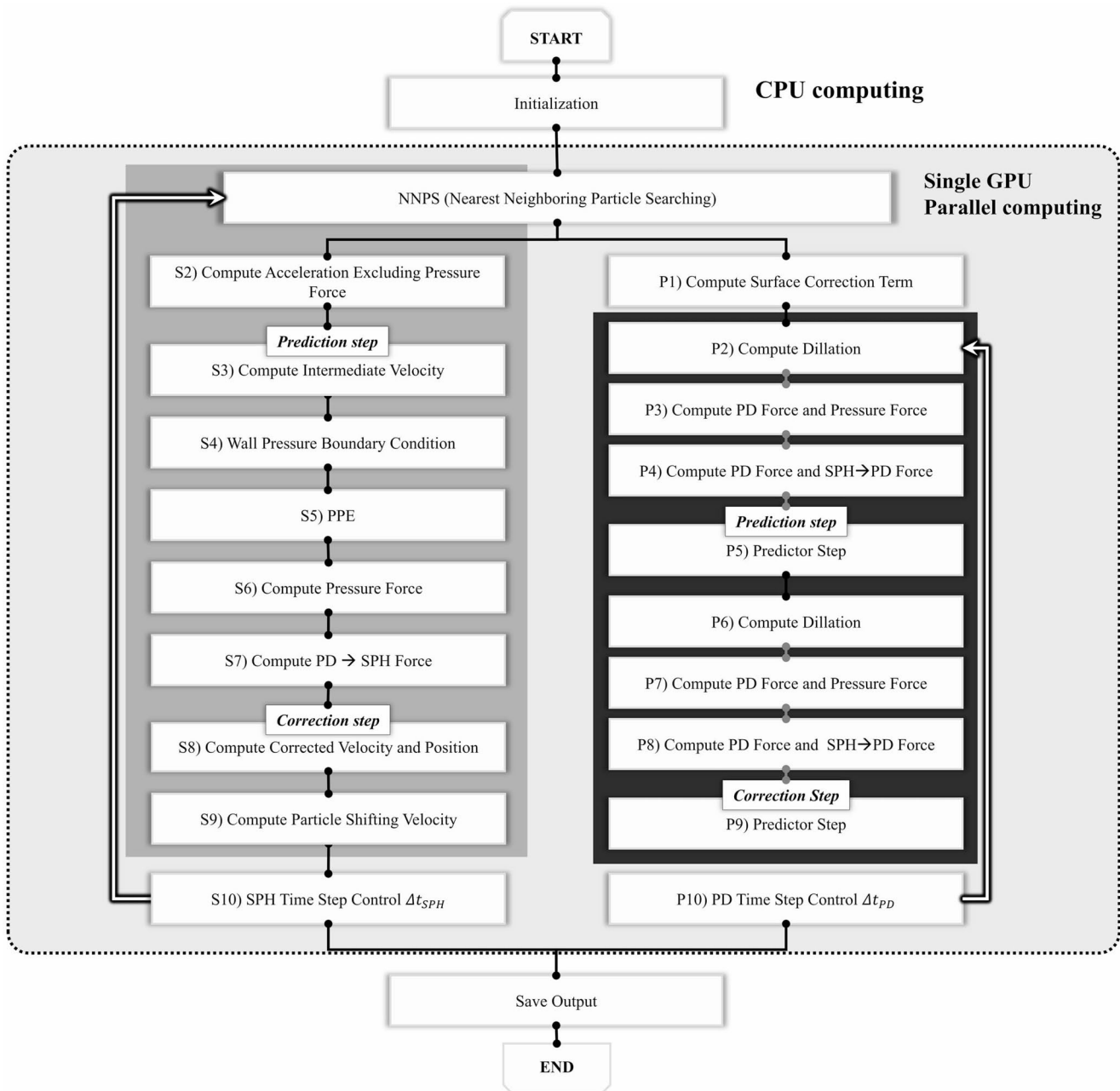


Fig. 6 Flowchart of the Coupled EISPH and OSPD

viscosity of water is defined as $\mu_0 = 0.01 \text{ Pa} \cdot \text{s}$. The initial pressure condition for the water is prescribed by using asymptotic solution [69]:

$$P(x, z) = \rho g(h - z) - \frac{8\rho gh}{\pi^2} \sum_{n=0}^{\infty} \left\{ \frac{1}{(2n+1)^2} e^{\frac{(2n+1)\pi z}{2H}} \cos\left(\frac{2n+1}{2h}\pi z\right) \right\} \quad (50)$$

Three different particle resolutions of coarse ($\Delta_{SPH} = h/20$), middle ($\Delta_{SPH} = h/30$), and fine

($\Delta_{SPH} = h/40$) are employed to evaluate the time evolution of the pressure load at selected pressure probes (P1, P2, P5, and P7). The influence of time step constraints is also investigated by testing two different CFL numbers, specifically 0.05 and 0.2.

As shown in Fig. 8, the simulation conducted with a CFL number of 0.05 at fine resolution yields a smooth and stable pressure field at selected time instances. No unphysical oscillations or discontinuities are observed near the obstruction or domain boundaries, demonstrating the robustness

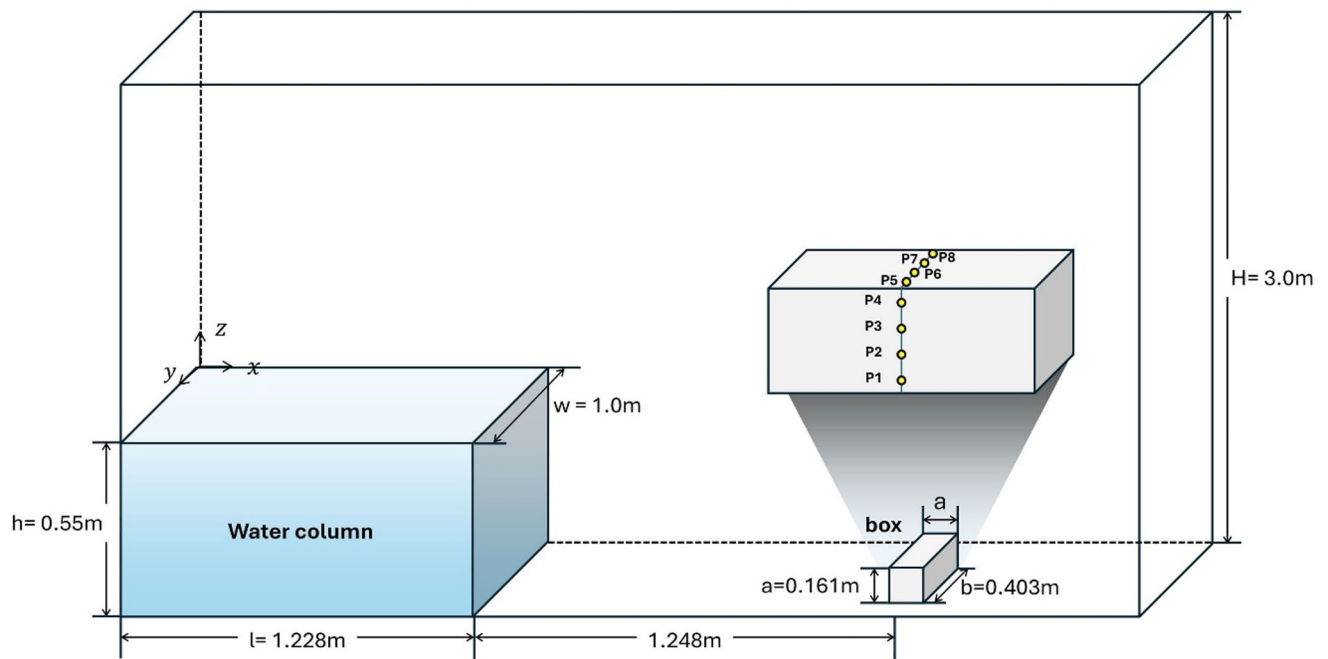


Fig. 7 Schematic Representation of Dam Break Problem with Obstacle

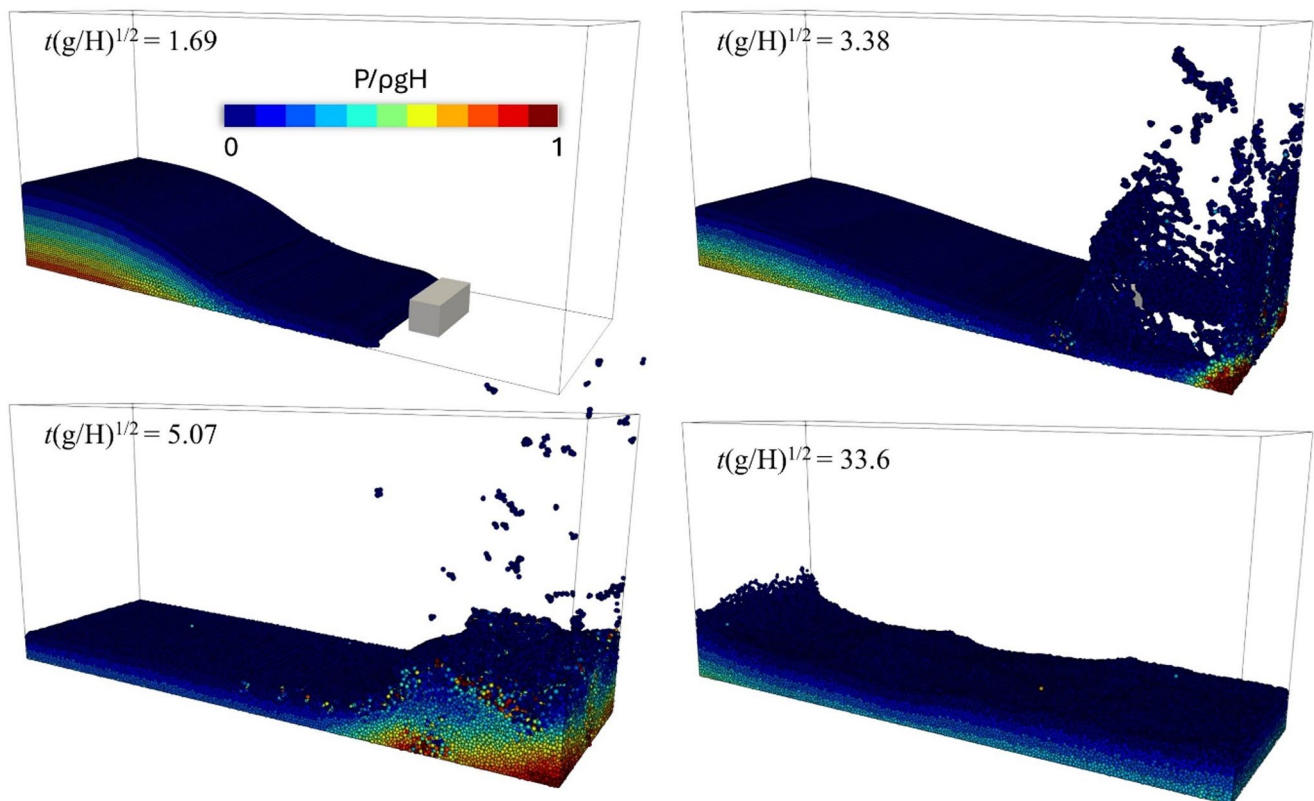


Fig. 8 Pressure Distribution Snapshots at Time Instants (CFL=0.05)

of the present SPH formulation under this time integration setting.

Furthermore, the time histories of pressure at the selected probes are compared with experimental data and numerical references, as presented in Fig. 9. The results show strong convergence behavior, with good agreement between numerical prediction and experiments [66] across different resolutions. In particular, the case with CFL = 0.05 shows improved accuracy in capturing the second pressure peak at probe P1.

Volume conservation has recently been recognized as an important topic in SPH research, particularly because its preservation over long-term simulations is crucial [70–73]. In addition, volume conservation was examined for both cases. Figure 10 shows the time evolution of the relative volume error in the dam-break problem. As shown in the figure, both cases exhibit good convergence, but the case with CFL = 0.05 demonstrates a noticeably better conservation

of volume. In our study, volume conservation is evaluated with the error volume defined as [72]:

$$\varepsilon_V = \frac{\sum_i V_{Ni}}{V_0} - 1 \quad (51)$$

where, the volume of particle i , denoted as V_{Ni} is defined based on the number density as:

$$V_{Ni} = \begin{cases} \frac{1}{\sum_j w_{ij}} & \forall i \notin \text{free surface } (\lambda_i \geq 0.6) \\ V_0 = \Delta_{SPH}^{dim} & \forall i \in \text{free surface } (\lambda_i < 0.6) \end{cases} \quad (52)$$

At several time instants, including a long-term simulation at $t(g/H)^{1/2} = 33.6$ the normalized volume computed based on number density (Eq. 51) and velocity divergence is presented in Fig. 11. As shown in the figure, local deviations appear in regions where wave breaking induces particle disorder, which temporarily degrades the divergence-free

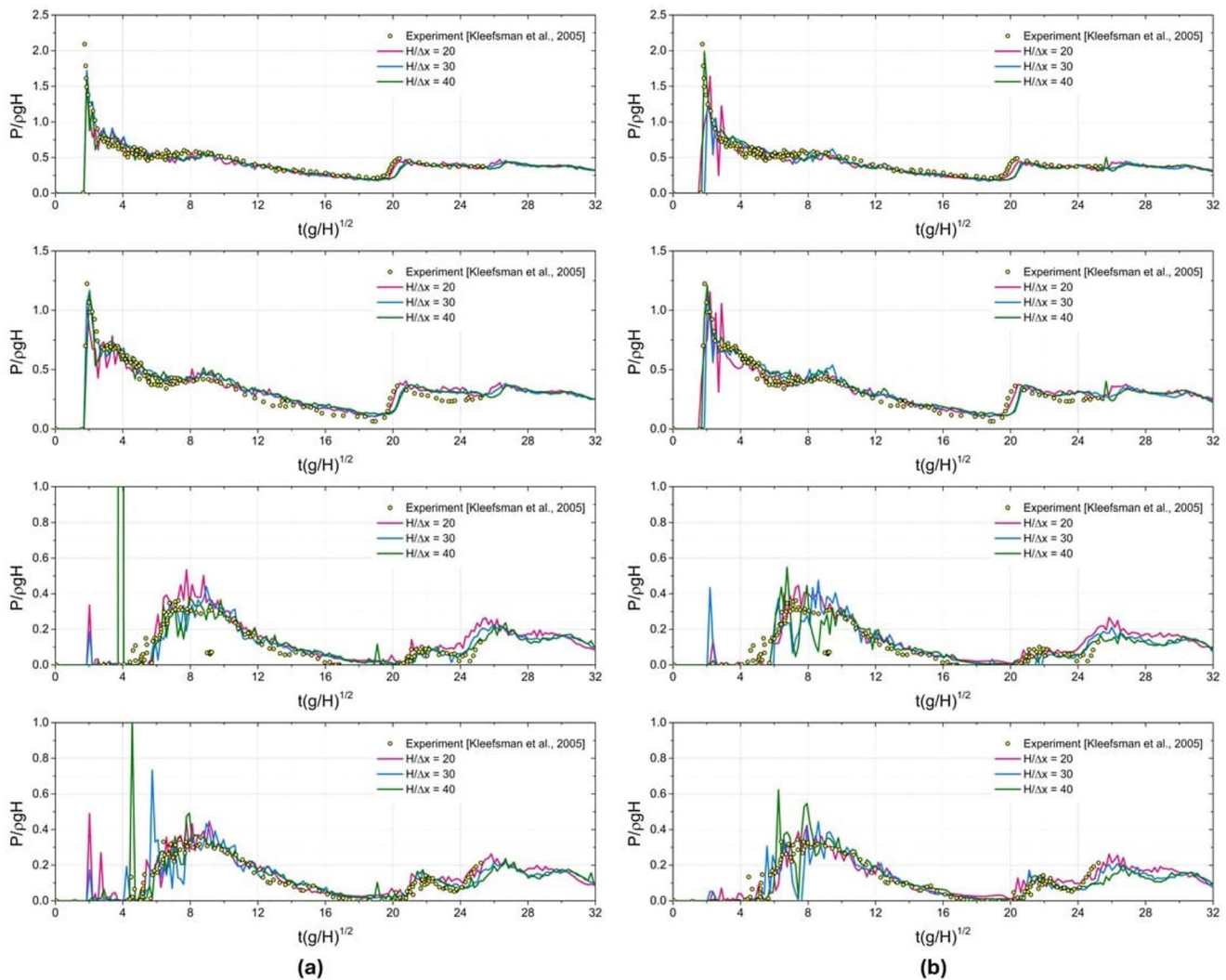


Fig. 9 Time evolution of pressure signals at probe positions comparing EISPH simulations and experiments. **a** CFL=0.2; **b** CFL=0.05. Lower CFL improves peak accuracy and temporal agreement

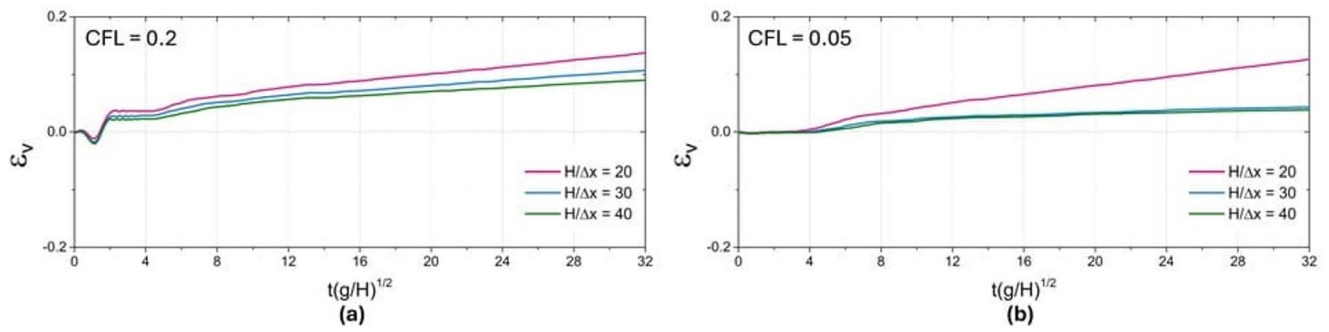


Fig. 10 Evolution of the relative volume error during the dam-break simulation with different spatial resolutions. **a** CFL=0.2; **b** CFL=0.05

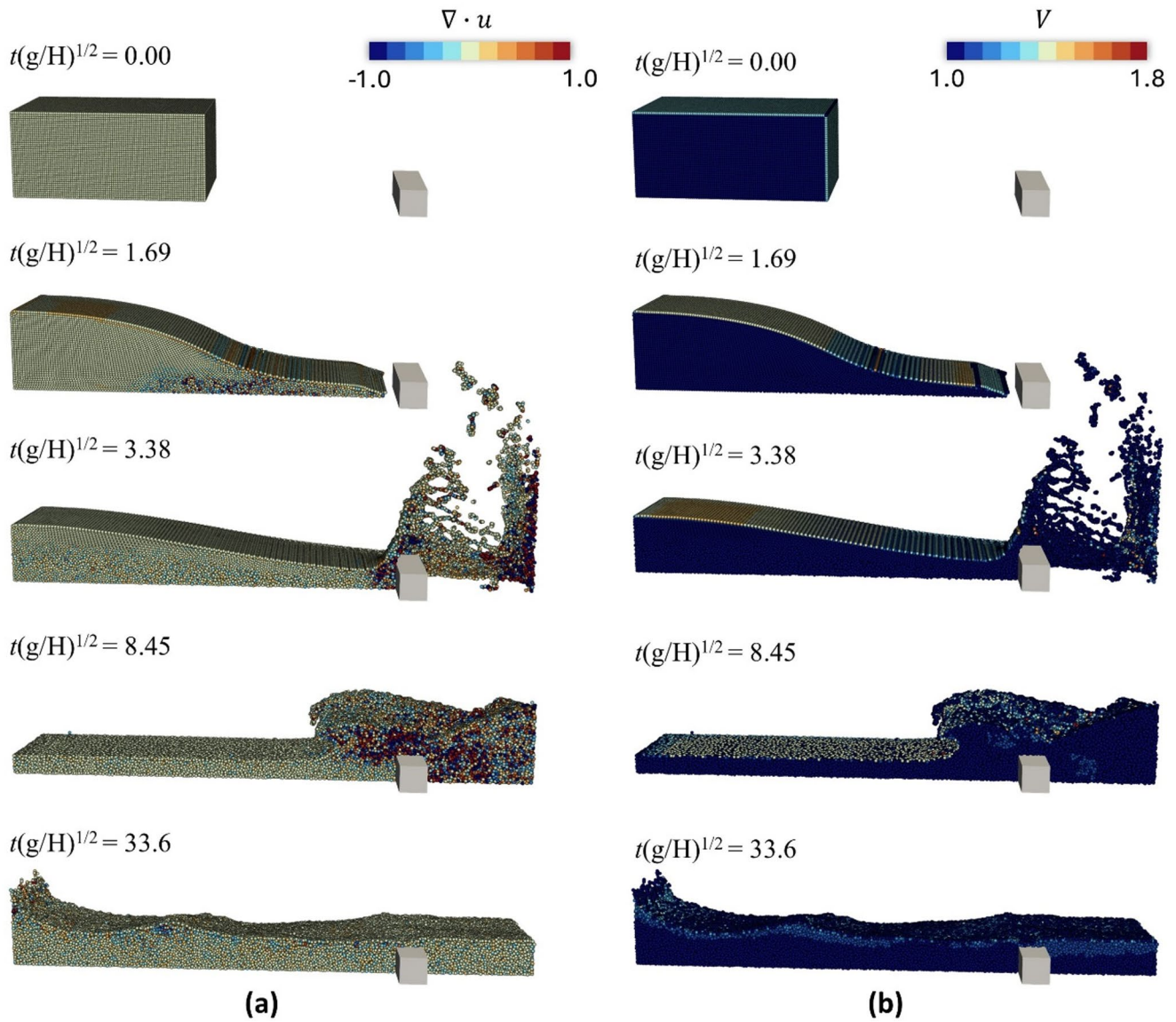


Fig. 11 Time evolution of **a** velocity divergence and **b** normalized volume error based on number density in the dam-break problem at CFL=0.05 with fine resolution case

condition of velocity. However, as the simulation progresses toward longer times, these deviations gradually diminish and approach zero overall. For the particle volume, although the free-surface region is prescribed as $\lambda_i < 0.6$, values between the bulk and the interface are not exactly unity due to particle number inconsistency. Nevertheless, the particle volume progressively recovers toward its original value over time, and no significant deterioration in volume conservation is observed, which demonstrates that the selected scheme exhibits acceptable long-term volume convergence. While various techniques have been developed to enhance volume conservation in both WCSPH and ISPH contexts, further dedicated investigations are still required in EISPH schemes [70–74].

4.2 Validation of OSPD model

To validate the robustness of the OSPD model employed in this study, two well-established benchmark problems are considered: the dynamic response of a freely oscillating cantilever beam, which focuses on elastic deformation [12, 75, 76], and the Kalthoff–Winkler experiment, which examines dynamic fracture propagation [77].

4.2.1 Cantilever beam

4.2.1.1 Experimental condition The dynamic response of a freely oscillating cantilever beam is analyzed to assess the accuracy of the OSPD model in capturing elastic deformation. Two scenarios are considered: one with relatively low stiffness and the other with high stiffness, representing a broad range of elastic behavior. As shown in Fig. 12a, the beam is fixed at the left end, which is attached to a rigid wall,

while all other boundaries are unconstrained. To enforce the clamped boundary condition at the left end, OSPD particles positioned at the wall are constrained to remain stationary throughout the simulation.

At the initial time step, the beam is assigned a transverse velocity profile in the y-direction based on the fundamental mode shape of a cantilever beam. The prescribed velocity field is given by [12, 76]:

$$u_y(x) = \xi c \frac{f(x)}{f(L)}, \quad \xi = 0.01, \quad c = \sqrt{\frac{K^s}{\rho^s}} \quad (53)$$

The $f(x)$ describes the normalized mode shape of the fundamental vibration mode as [12, 76]:

$$f(x) = (\cos(k_w L) + \cosh(k_w L)) (\cosh(k_w x) - \cos(k_w x)) + (\sin(k_w L) - \sinh(k_w L)) (\sinh(k_w x) - \sin(k_w x)) \quad (54)$$

The analytical solution for the displacement at any point on the beam is given by:

$$y(t) = A_{max} \sin(\omega t) \quad (55)$$

where the amplitude A_{max} and angular frequency ω are defined as:

$$A_{max} = \frac{u_y(x=L)}{\omega}, \quad \omega = k_w^2 \sqrt{\frac{E^s I_y}{\rho^s S}} \quad (56)$$

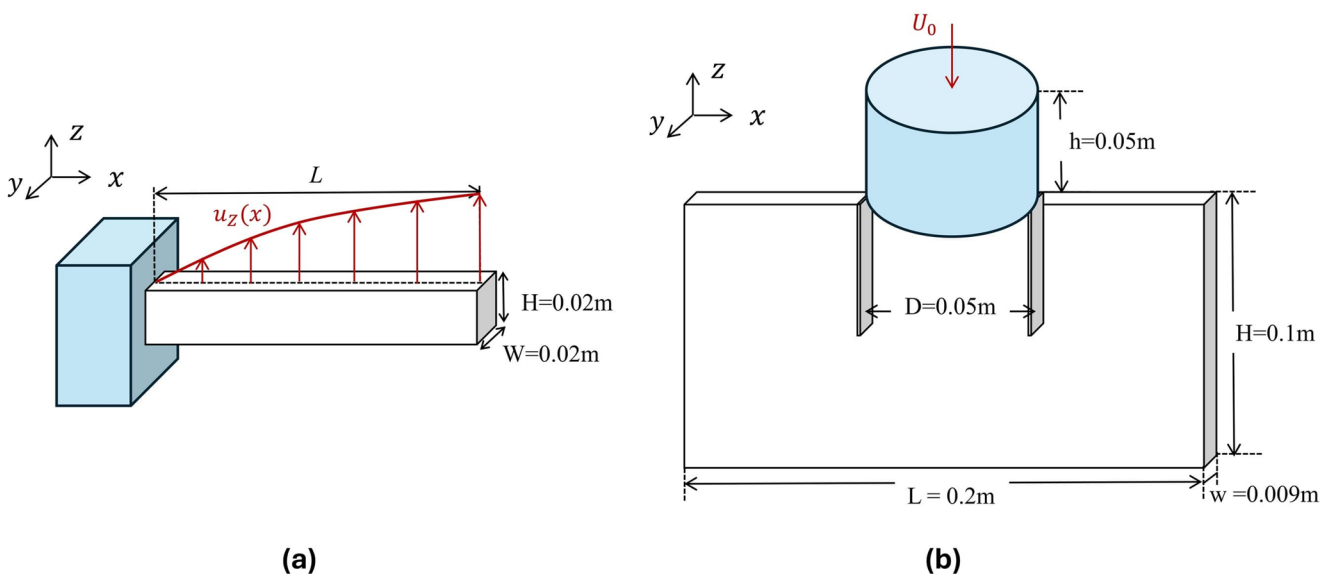


Fig. 12 Schematic Representation of Benchmark Problems: (a) Freely Oscillating Cantilever beam and (b) Kalthoff–Winkler Experiment

Fig. 13 Comparison of Displacement Response using Different Influence Functions without Correction at Middle Resolution ($\Delta_{PD} = L/100$)

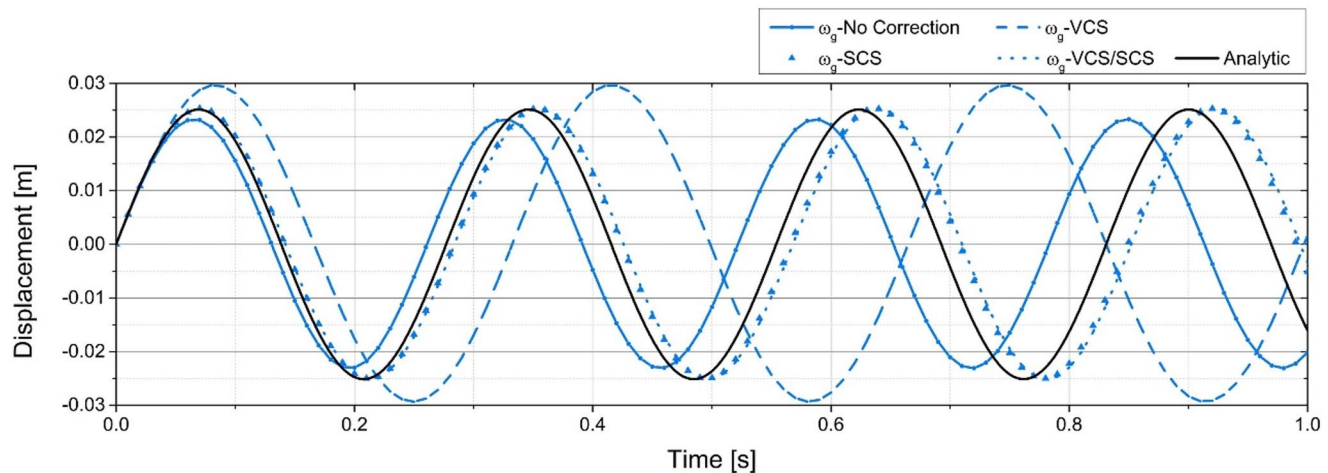
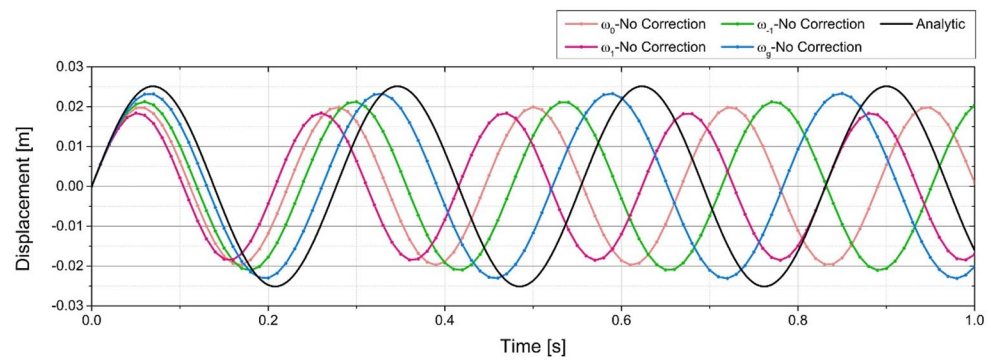


Fig. 14 Effect of Correction Schemes (VCS, SCS) with Influence Function ω_g at Middle Resolution ($\Delta_{PD} = L/100$)

Here, E^s is the Young's modulus, $S(=WH)$ is the cross-sectional area, and $I_y(=WH^3/12)$ is the second moment of area about the neutral axis, where W and H are the width and height of the beam, respectively. For the fundamental mode of vibration, the wave number k_w satisfies $k_w L = 1.875$.

4.2.1.2 Numerical results To evaluate the accuracy of the OSPD model in the elastic regime, we first focus on the relatively elastic case ($E^s = 2\text{MPa}$, $L = 0.2\text{m}$). Three levels of spatial resolution are considered, namely coarse ($\Delta_{PD} = L/50$), middle ($\Delta_{PD} = L/100$), and fine ($\Delta_{PD} = L/200$), to examine the sensitivity of the model to discretization.

For the middle-resolution case ($\Delta_{PD} = L/100$), four different influence functions (ω_{-1} , ω_0 , ω_1 , ω_g) are tested to investigate their impact on the dynamic response. As shown in Fig. 13, all influence functions lead to noticeable deviations from the analytical solution, indicating that the choice of influence function alone is insufficient to guarantee accurate results in OSPD model.

The Figure 14 compares the performance of correction schemes using the ω_g at the middle resolution. Results

without correction, as well as with VCS, SCS and the combined VCS-SCS approach (VCS/SCS) are presented. While VCS reduces local numerical errors to some extent, its effect is limited in improving global accuracy. In contrast, SCS significantly improves the agreement with the analytical solution. Interestingly, the combination of VCS and SCS (VCS/SCS) does not result in noticeable improvement beyond SCS alone, as shown in Fig. 14.

To further assess the effect of influence function choice in conjunction with correction schemes, Fig. 15 presents the results obtained using different influence functions with VCS/SCS applied. It is observed that the incorporation of correction schemes significantly improves the accuracy of the elastic deformation response across all influence functions tested.

The resolution study is performed using four influence functions (ω_{-1} , ω_0 , ω_1 , ω_g) corrected with VCS/SCS scheme, evaluated across three spatial resolutions: coarse, middle and fine. The time histories of displacement are compared to the analytical solution, as shown in Fig. 16. The results confirm that the present model exhibits good convergence behavior with increasing resolution, and that the corrected formulations consistently improve accuracy across all influence functions tested.

Fig. 15 Comparison of Displacement Response using Different Influence Functions with VCS/SCS scheme at Middle Resolution ($\Delta PD = L/100$)

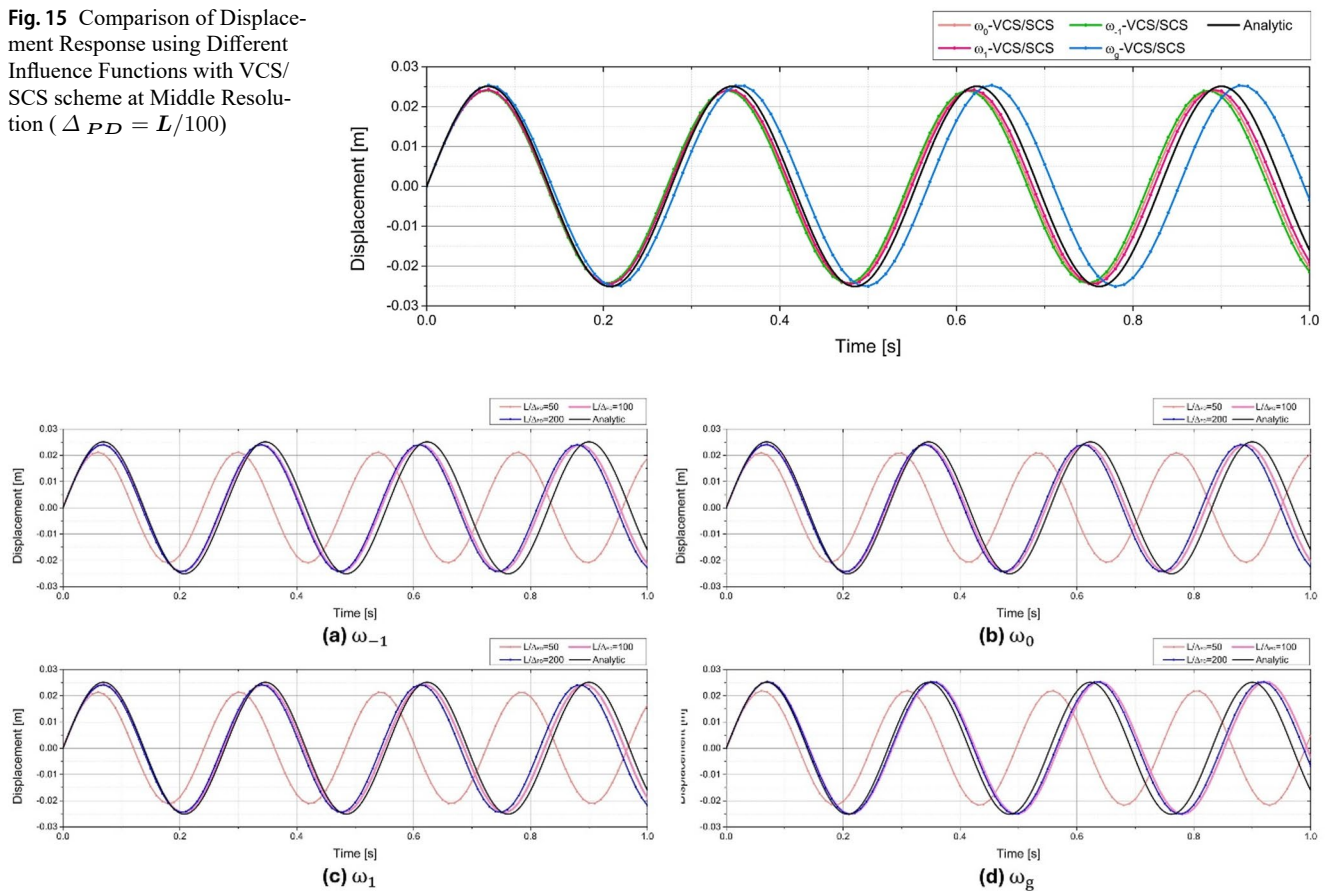


Fig. 16 Time Histories of Displacement Response at Coarse, Middle, and Fine Resolutions using ω_{-1} , ω_0 , ω_1 , ω_g with VCS / SCS Correction

In addition, Fig. 17 shows the dilatation results obtained from different time integration schemes under the low-resolution setting with the Gaussian influence function. The results from the Euler and predictor–corrector schemes are compared. The Euler scheme produces checkerboard-like patterns in the dilatation field, indicating the presence of nonphysical oscillations and numerical instability. In contrast, the predictor–corrector scheme suppresses these artifacts and produces a smoother and more stable dilatation response.

The Figure 18 displays the dilatation field obtained at fine resolution using the Gaussian influence function (ω_g). As shown in the figure, the dilatation field remains smooth and stable, indicating good numerical consistency and the absence of nonphysical artifacts.

Based on the results from the elastic regime, we next evaluate the model performance for a stiff material case ($E^s = 2\text{GPa}$, $L = 0.5\text{m}$). Similar to previous case, simulations are conducted at three spatial resolutions to evaluate the numerical behavior under high stiffness conditions. Figure 19 presents the time histories of displacement in the stiff case evaluated at three spatial resolutions. The results

demonstrate good convergence between the analytical result and numerical results with increasing resolution.

Figure 20 shows the dilatation field obtained with the Gaussian influence function (ω_g) at fine resolution ($\Delta PD = L/200$). The field remains smooth and free from numerical artifacts, confirming the stability and consistency of the OSPD model even in the high-stiffness regime.

4.2.2 Kalthoff-Winkler experiment

4.2.2.1 Experimental condition To further validate the accuracy and robustness of the OSPD formulation, the well-known Kalthoff–Winkler impact experiment is numerically reproduced [24, 78]. As illustrated in Fig. 12b, a rectangular 18Ni1900 steel plate containing two symmetrically placed pre-existing cracks located at a 0.025 m distance from the centerline is subjected to high-speed vertical impact by a cylindrical projectile. The mass of projectile is $M = 1.57\text{kg}$. The material properties of plate are used as $\rho^s = 7800\text{kg/m}^3$, $\nu^s = 0.25$, and $E^s = 190\text{GPa}$, $G_0 = 69\text{kJ/m}^2$ as in the experiment. The geometric dimensions of the plate and projectile follow the standard configuration of the original

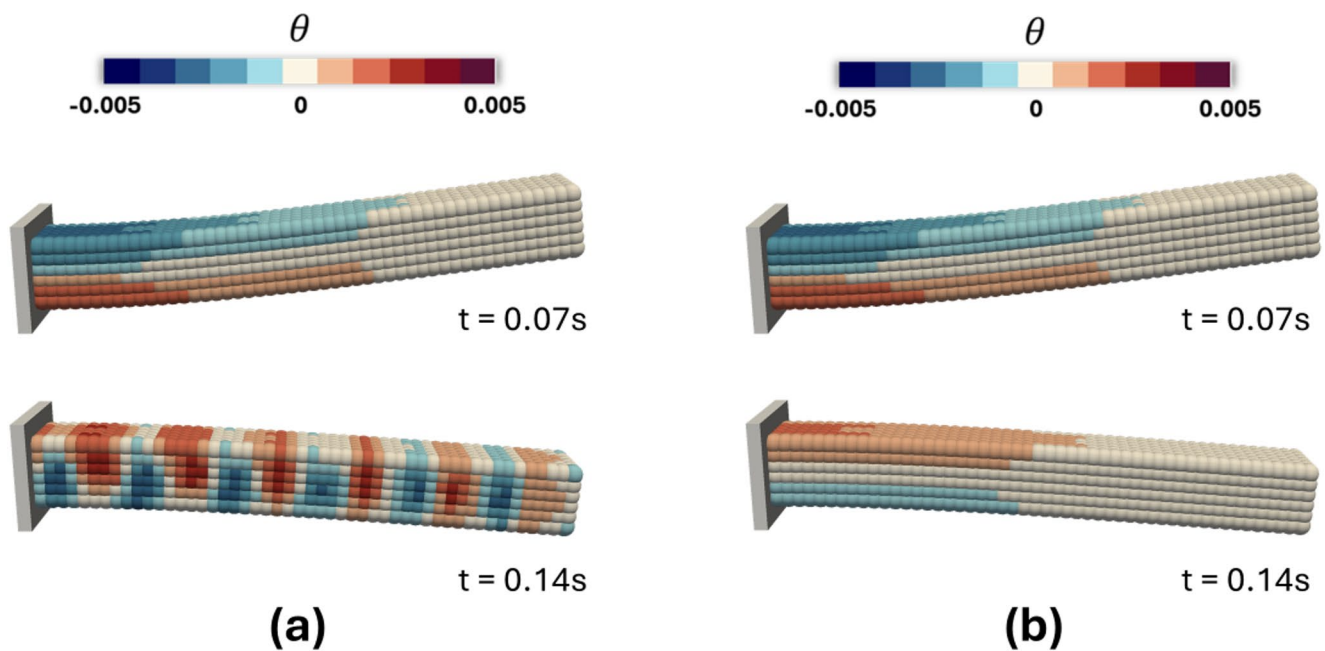


Fig. 17 Comparison of Dilatation Fields using (a) Euler and (b) Predictor-Corrector Time Schemes at 0.11s with Coarse Resolution ($\Delta_{PD} = L/50$)

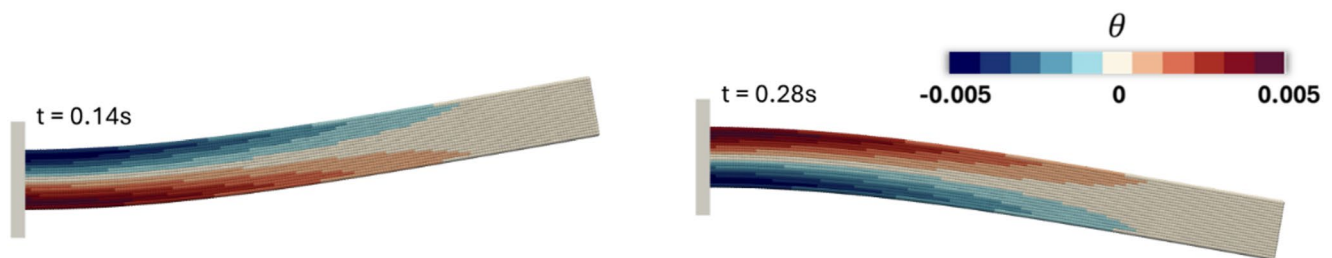


Fig. 18 Dilatation field at fine resolution ($\Delta_{PD} = L/200$) using the Gaussian influence function (ω_g)

Fig. 19 Time Histories of Displacement in the Stiff Case at Coarse, Middle, and Fine Resolutions using ω_g with VCS / SCS Correction

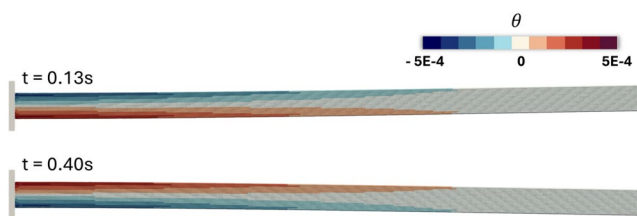
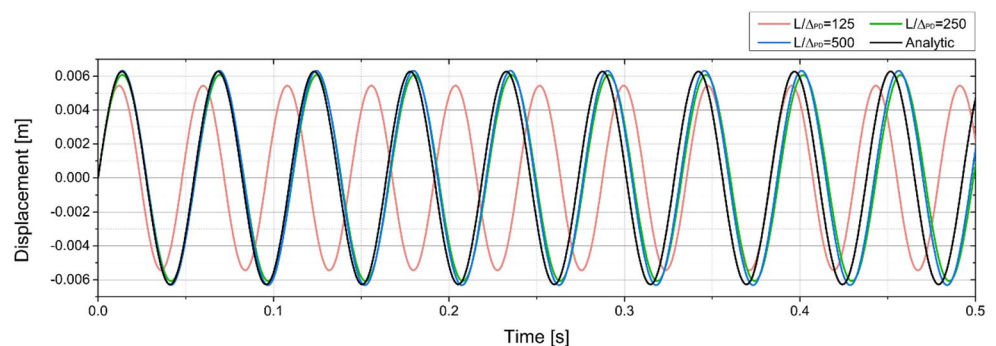


Fig. 20 Dilatation field in the Stiff Case at fine resolution ($\Delta_{PD} = L/200$) using the Gaussian influence function (ω_g)

experiment as in the Fig. 12b. The particle resolution is set as same as the previous literature [24].

4.2.2.2 Numerical results The damage field and the location of the propagated crack using two different influence functions (ω_{-1} and ω_0) are presented in Fig. 21. Preliminary tests indicated that the choice of influence function resulted in only minor variations in local accuracy prior

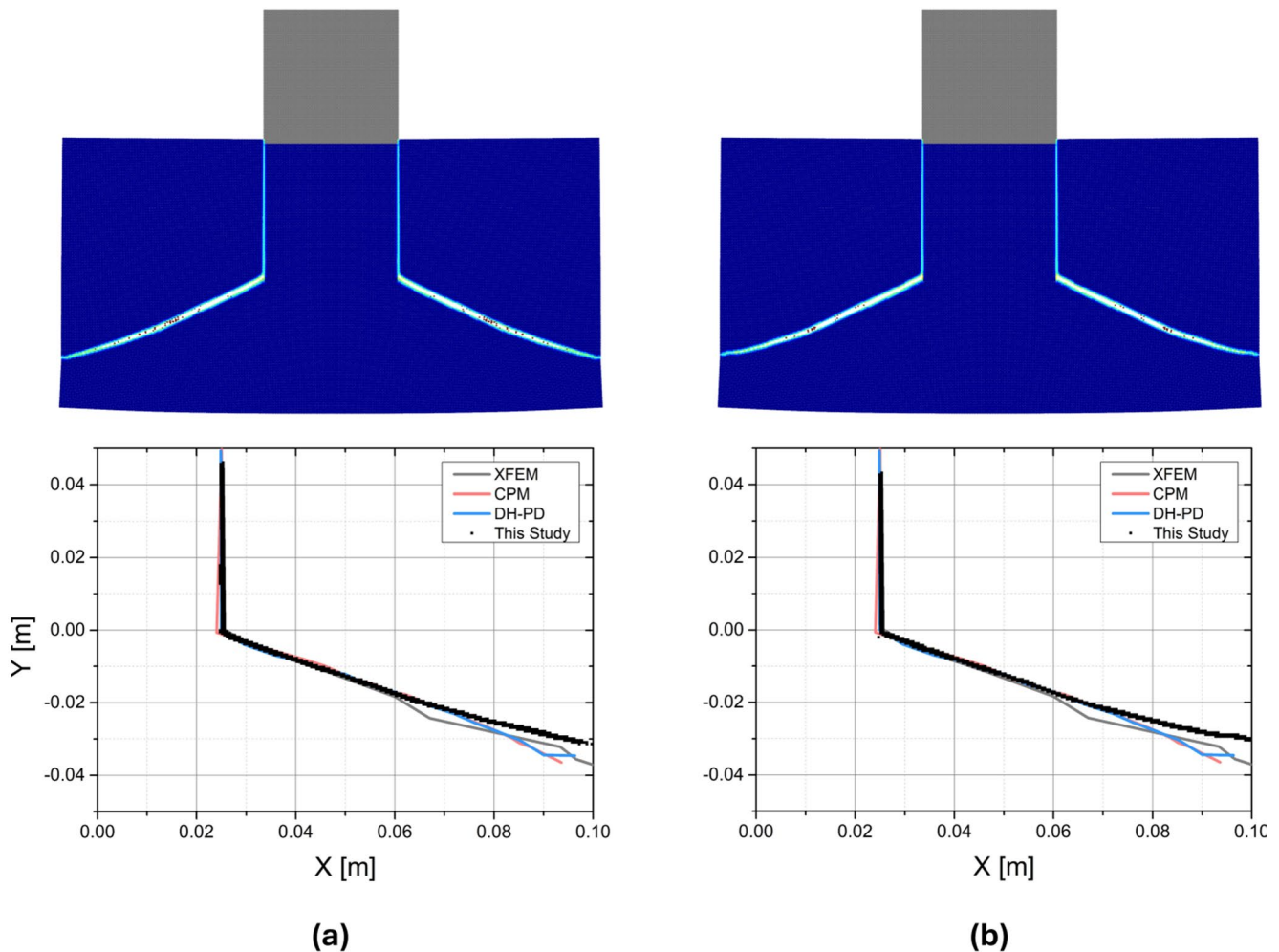


Fig. 21 Damage field and propagated crack at $t = 8 \mu s$, compared with results from previous studies using (a) ω_{-1} and (b) ω_0

to fracture. Therefore, representative influence functions widely adopted in previous studies were selected to ensure both practicality and consistency in the subsequent fracture analyses. As shown in the figure, the predicted crack propagation paths are generally consistent with results from previous studies based on different numerical methods employing with CPM [79], XFEM [80], and Dual Horizon PD [24]. In particular, the crack trajectories near the pre-existing flaws exhibit strong agreement across methods. However, slight differences are observed in the crack tip morphology, which vary not only between numerical approaches but also depending on the choice of influence function. However, the differences are minor and remain within a comparable range across the methods.

The time evolution of the damage field and velocity field using the ω_{-1} influence function is presented in Fig. 22. As shown in the figure, the velocity field remains stable and continuous throughout the simulation, indicating good numerical stability and physical consistency.

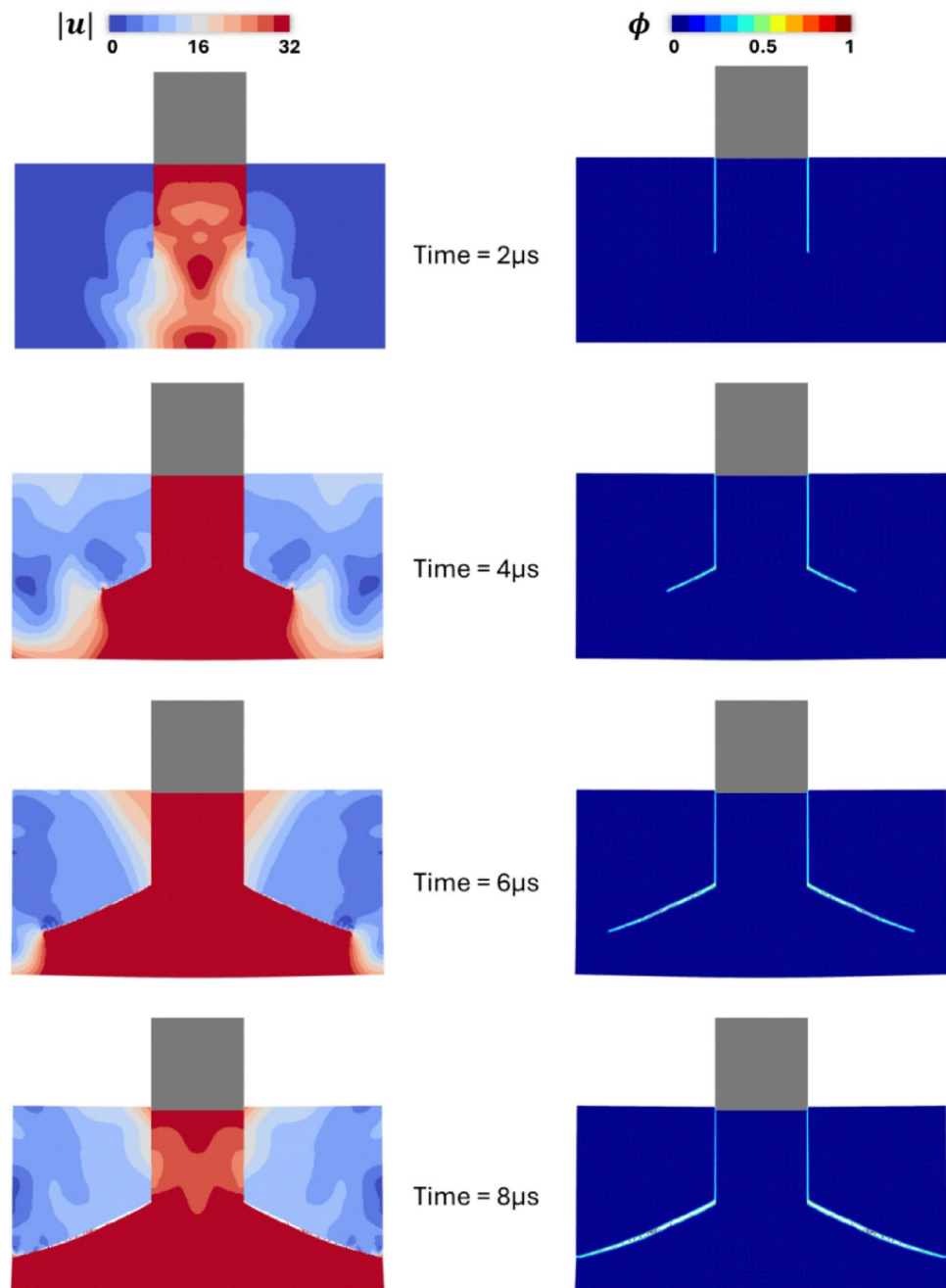
4.3 Validation of EISPH-OSPD coupling

In this section, the coupled EISPH and OSPD scheme is validated through a series of benchmark problems, which are described in Fig. 23. These include (a) dam-break flow through an elastic gate, (b) dam-break flow involving an elastic sluice gate, and (c) dam-break flow involving an elastic baffle fixed to the bottom wall. Figure 23 illustrates the summary of the key geometric information of each benchmark problem. In this study, the particle resolutions of the SPH and PD domains are set to be identical. Accordingly, the initial particle spacing is denoted as $\Delta x = \Delta_{SPH} = \Delta_{PD}$. In addition, the most simple constant influence function (ω_0) is adopted for the PD computation.

4.3.1 Dam-break flow through an elastic gate

4.3.1.1 Experimental condition The benchmark problem involving dam-break flow through an elastic gate is one of the most well-established test cases for fluid–structure inter-

Fig. 22 Time Histories of the Velocity Field and Damage Field using the ω_{-1} Influence Function



action involving free-surface flow [11]. It focuses on the deformation of an elastic plate subjected to transient water pressure. The physical configuration follows the experimental setup reported in the literature and is summarized in Fig. 23a.

The fluid properties are defined as density $\rho = 1000 \text{ kg/m}^3$, and dynamic viscosity is set as $\mu = 0.01 \text{ Pa} \cdot \text{s}$, which represents a reasonable trade-off between physical fidelity and computational robustness in lots of previous literatures. The material properties of the

elastic gate have varied across different studies as presented in Table 1. In this work, the properties are adopted from a previous study that employed a robust ISPH-SPH solver as its foundation [81]. The elastic plate, made of rubber in the experiment, is modeled with a density $\rho^s = 1100 \text{ kg/m}^3$, Young's modulus $E^s = 12 \text{ MPa}$, and Poisson's ratio $\nu^s = 0.4$ [81]. The initial pressure distribution in the water column is prescribed using an asymptotic solution [69]. Two particle resolutions are considered in this study: $\Delta x = a/5$ and $\Delta x = a/7.5$, where a is the thickness of the elastic gate displayed in Fig. 23a.

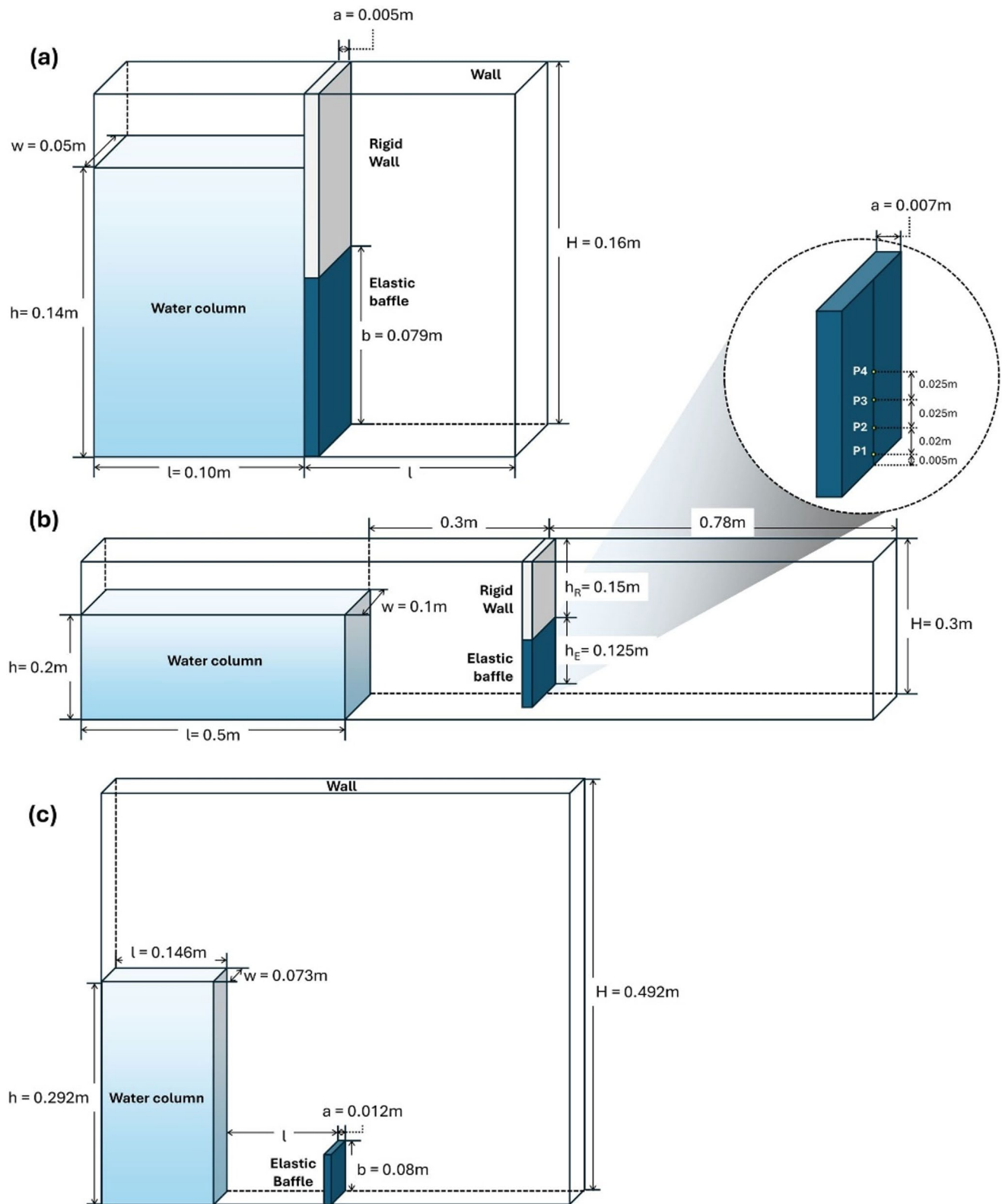


Fig. 23 Geometries of Benchmark Cases for FSI Validation: (a) Dam-Break Flow Through an Elastic Gate, (b) Dam-Break Flow with an Elastic Sluice Gate, and (c) Dam-Break Flow Involving an Elastic Baffle Fixed to the Bottom Wall (pressure probe)

Table 1 List of material properties used in previous literature and the present study

	ρ^s [kg/m ³]	E^s [MPa]	ν^s
WCSPH-SPH [11]	1100	12	0.4
PD-WCSPH [21]	1100	12	0.25
SPH-FEM [82]	1100	12	0.4
ISPH-SPH [81]	1100	5.5	0.45
This Study	1100	12	0.4

4.3.1.2 Numerical results Figure 24 illustrates a sequence of time snapshots showing the water flow and the corresponding deformation of the elastic gate. In the visualization, the water column is colored by the pressure field, while the elastic gate is mapped by the dilatation field to represent its deformation behavior. At the beginning of the simulation, the water column is held at a specified height. Once the experiment starts, the hydrostatic pressure exerts a force on the elastic gate, causing it to deform and allowing water to flow through the opening. As the water level decreases due to this flow, the pressure acting on the gate is reduced, and the elastic plate gradually attempts to return to its original configuration.

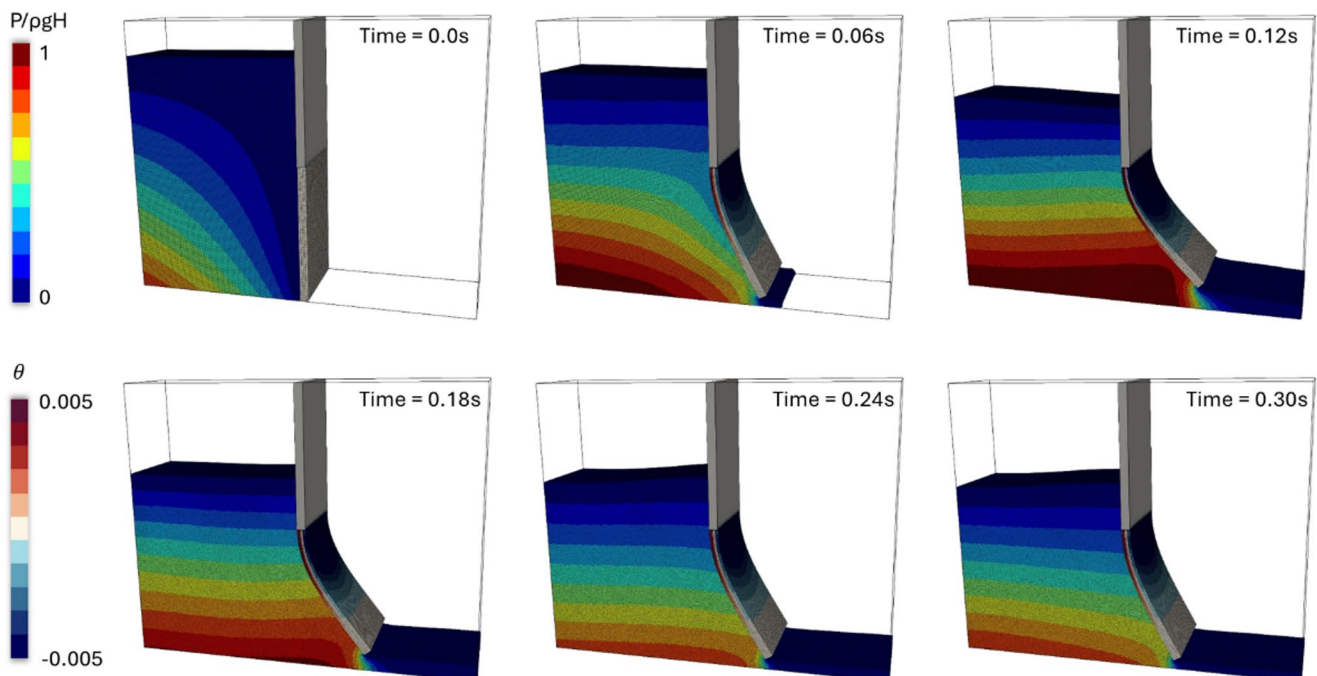
The computing time at fine resolution case ($\Delta x = a/7.5$) is compared between uniform time stepping and sub-cycling in the Fig. 25. In the uniform scheme, the global timestep is constrained by the relatively small timestep of PD particles. As shown in the figure, when the PD solver advances to about 20,000 steps, the total computing time of the uniform

scheme becomes more than five times longer than that of the sub-cycling scheme. The efficiency gain depends on the relative number of SPH / PD particles, and the benefit might become more significant in this type of FSI problem when the number of PD particles is relatively small.

The deformation of the elastic gate is quantitatively evaluated by comparing the displacement of its bottom tip with experimental data and other numerical results performed by different numerical methods including WCSPH-SPH [11], WCSPH-PD [21], SPH-FEM [82], and ISPH-SPH [81]. Specifically, both the horizontal (x) displacement is tracked over time. As shown in Fig. 26, the case with coarse particle resolution underpredicts the deformation compared to the reference numerical results using the same material properties. In contrast, the finer resolution case yields results that closely match those reported in the reference study. Overall, the present results fall within the range of existing numerical predictions and demonstrate consistent and acceptable agreement.

4.3.2 Dam-break flow involving an elastic sluice gate

4.3.2.1 Experimental condition As the second benchmark case, a dam-break flow involving an elastic sluice gate is considered and compared with experiment and numerical results [83]. In this problem, the water column collapse and flows toward the gate, initiating dynamic interaction. As the

**Fig. 24** Time Evolution of Pressure Field of SPH Fluid Particles and Dilatation Fields of PD Structure Particles in Dam-Break Flow Through an Elastic Gate at $\Delta x = a/7.5$

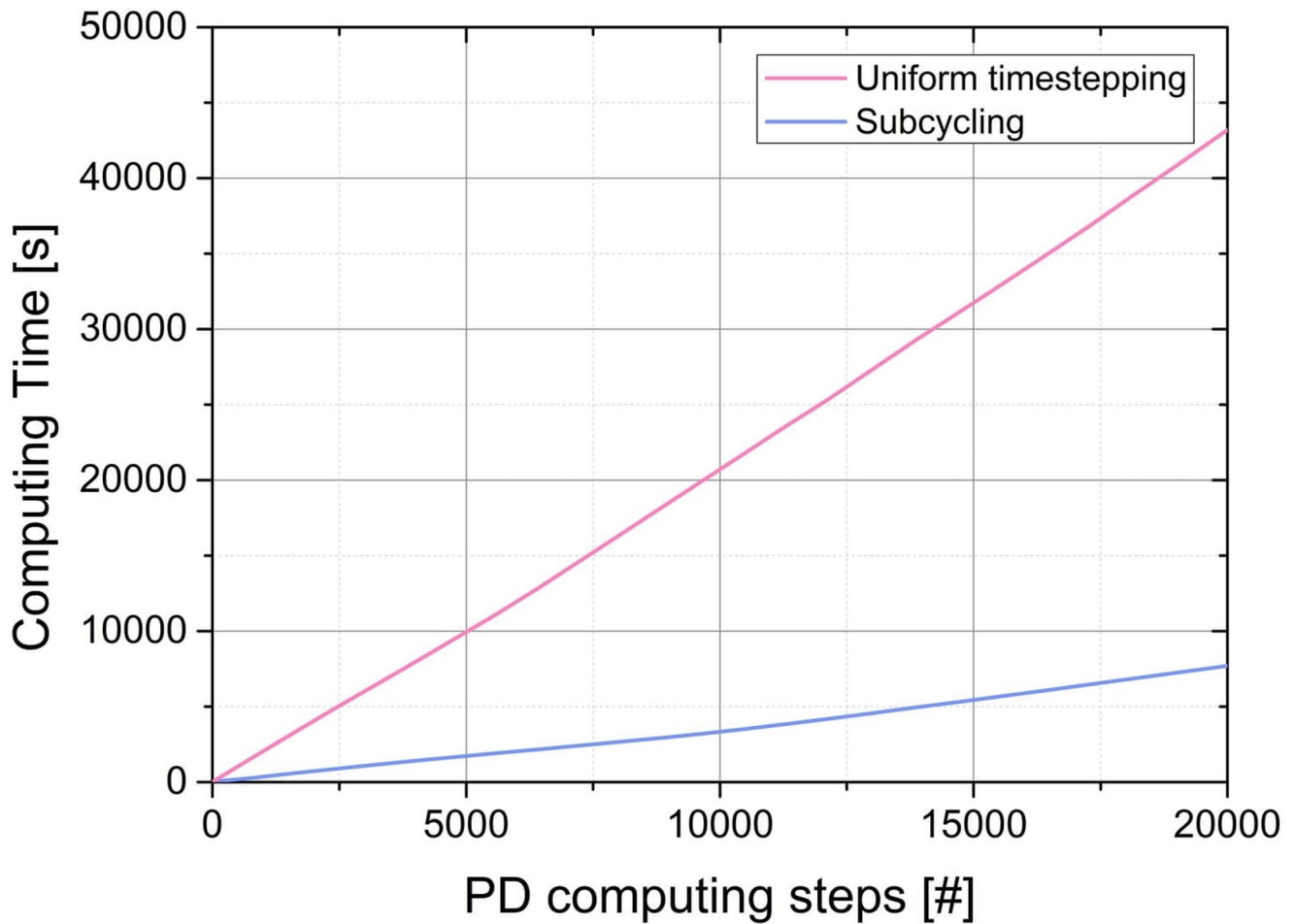


Fig. 25 Comparison of Computing Time Between Uniform Time Stepping and Sub-Cycling in the Dam-Break Flow Through an Elastic Gate at Fine Resolution at $\Delta x = a/7.5$

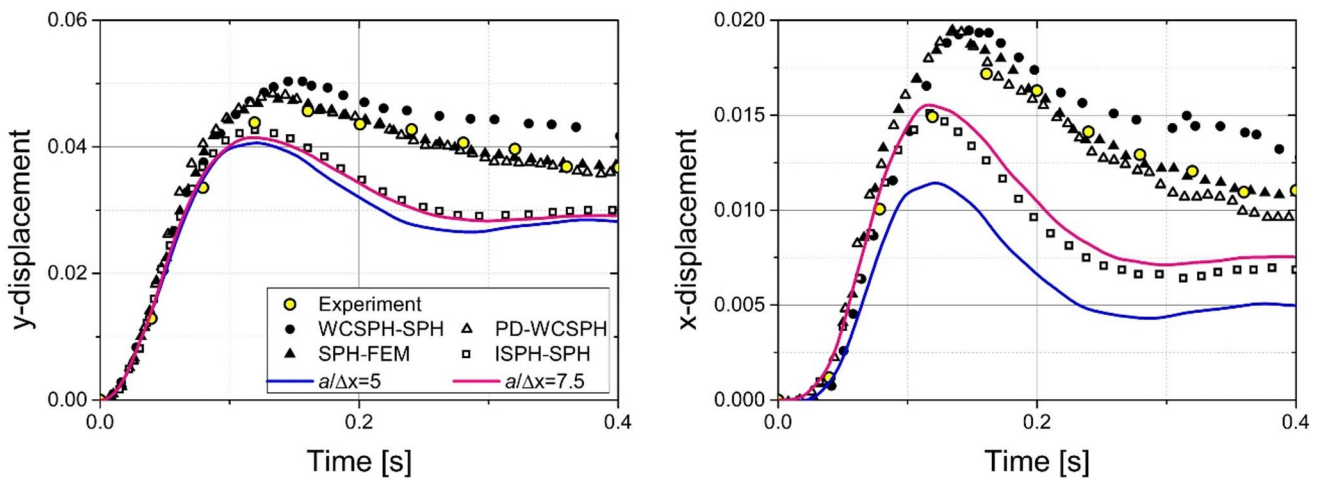


Fig. 26 Horizontal Displacement of the Gate Tip Compared with Experimental and Numerical Results

free surface grazes the gate, deformation of the elastic structure is induced due to the resulting hydrodynamic pressure.

The fluid properties are defined as density $\rho = 1000 \text{ kg/m}^3$, and dynamic viscosity $\mu = 0.01 \text{ Pa} \cdot \text{s}$. The elastic plate, made of rubber-like in the experiment, is modeled with a density $\rho^s = 1250 \text{ kg/m}^3$, Young's modulus $E^s = 4 \text{ MPa}$, and Poisson's ratio $\nu^s = 0.4$. The initial pressure distribution in the water column is prescribed using an asymptotic solution. Initial particle distance is set as $\Delta x = a/5$, where a is the thickness of the elastic baffle as in Fig. 23b.

4.3.2.2 Numerical results As illustrated in Fig. 27, the simulation qualitatively reproduces the overall interface motion and structural deformation in good agreement with the experimental observations [83]. The pressure field within the fluid remains stable throughout the simulation, while the dilatation field in the structure exhibits smooth and physically consistent behavior, demonstrating the robustness and reliability of the proposed numerical scheme. At $t = 0.3 \text{ s}$, a thin stream of water is observed following the contour of the elastic baffle as the flow contacts its tip. As the water continues to impinge, it develops into a wave, and this dynamic evolution of the free surface is well captured in the simulation results.

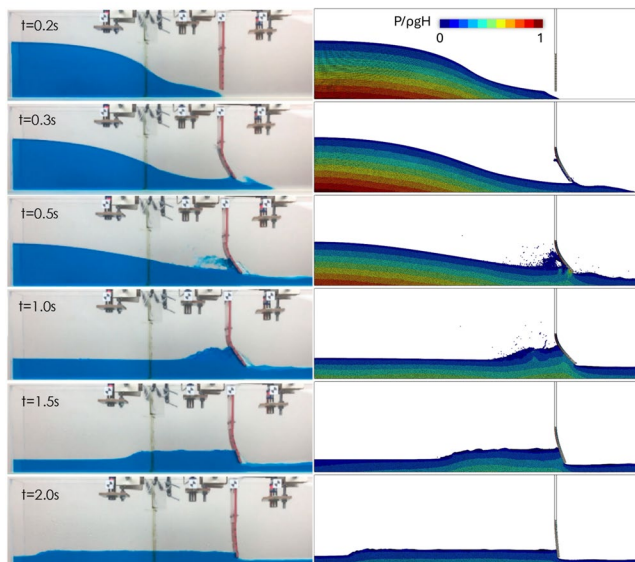


Fig. 27 Time Snapshots of Dam-Break Flow Interacting with an Elastic Sluice Gate Showing Pressure Field of SPH Fluid Particles and Dilatation Fields of PD Structure Particles at $\Delta x = a/5$. Experimental data overlaid for comparison are reproduced from Fig. 14 in Yilmaz et al. (2021) [1]. (permission requested).

Although PD does not require stress tensors for computing internal forces as opposed to classical continuum mechanics, the stress field can still be reconstructed from pairwise force densities in a post-processing step. This reconstructed stress field offers valuable insight into the structural response and local stress distribution. The stress tensor at each material point can be computed similarly as the virial theorem in molecular dynamics [84, 85]:

$$\sigma_A = \frac{1}{2V_A} \sum_{B \in H_A} (\mathbf{y}_B - \mathbf{y}_A) \otimes \mathbf{F}_{AB} \cdot \mathbf{F}_{AB} = (\mathbf{t}_{AB} - \mathbf{t}_{BA}) V_A V_B \quad (57)$$

For the example shown in Fig. 25, the computed stress field is smooth and free from spurious oscillations, further demonstrating the robustness of the EISPH-OSPD formulation.

In Fig. 28, the reconstructed stress field is displayed. For clarity, only the free surface of the fluid is included using a contour representation. As shown, the stress distribution is remarkably smooth and exhibits a symmetric pattern, demonstrating the numerical stability of the proposed method.

To quantitatively assess the accuracy of the fluid–structure interaction, time histories of the water surface elevation are recorded at three probe points (H1, H2, and H3, each at 0.35 m, 0.6 m, 0.95 m from left wall), while the horizontal displacements of the elastic gate are tracked at structural markers (P1, P2, P3, and P4 in Fig. 23b). These results are compared against experimental data [83]. As shown in Fig. 29, the present method captures consistent trends in both the fluid and structural responses, confirming its reliability in modeling complex FSI phenomena.

4.3.3 Dam-break flow interacting with an elastic baffle attached to the bottom wall

4.3.3.1 Experimental condition As the third benchmark case, a dam-break flow interacting with an elastic baffle attached to the bottom wall is considered, which has been studied by many literatures [21]. In this problem, a water column initially held at rest collapses and flows toward the elastic baffle, exerting hydrodynamic pressure upon impact. The baffle undergoes significant deformation due to the fluid loading, and its subsequent recovery motion leads to complex flow–structure interactions. The dynamic coupling between the deforming baffle and the unsteady fluid motion results in rich and nonlinear flow behavior.

In this study, the deformation of the elastic baffle under such transient hydrodynamic loading is captured and compared with available reference simulations to validate the predictive capability of the proposed method. The fluid properties are defined as density $\rho = 1000 \text{ kg/m}^3$, and dynamic viscosity $\mu = 0.01 \text{ Pa} \cdot \text{s}$. The elastic plate is

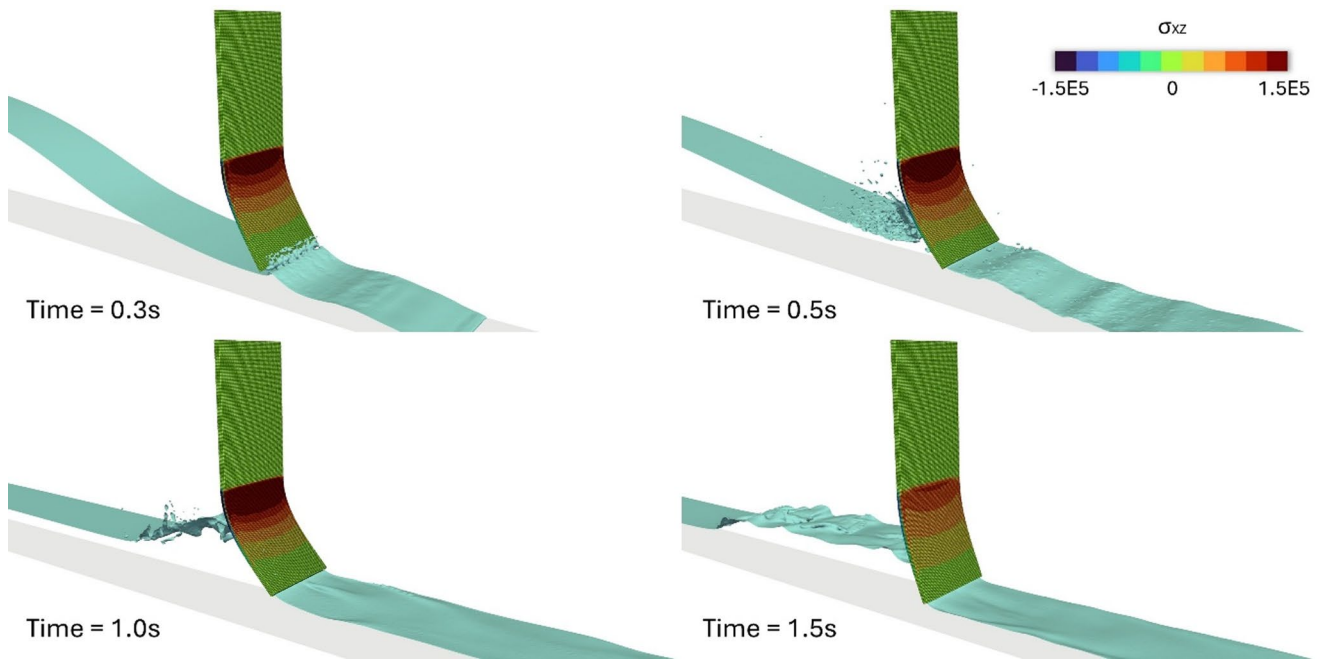


Fig. 28 Reconstructed Stress Field of the PD Structure Under Dam-Break Flow Interacting with an Elastic Sluice Gate

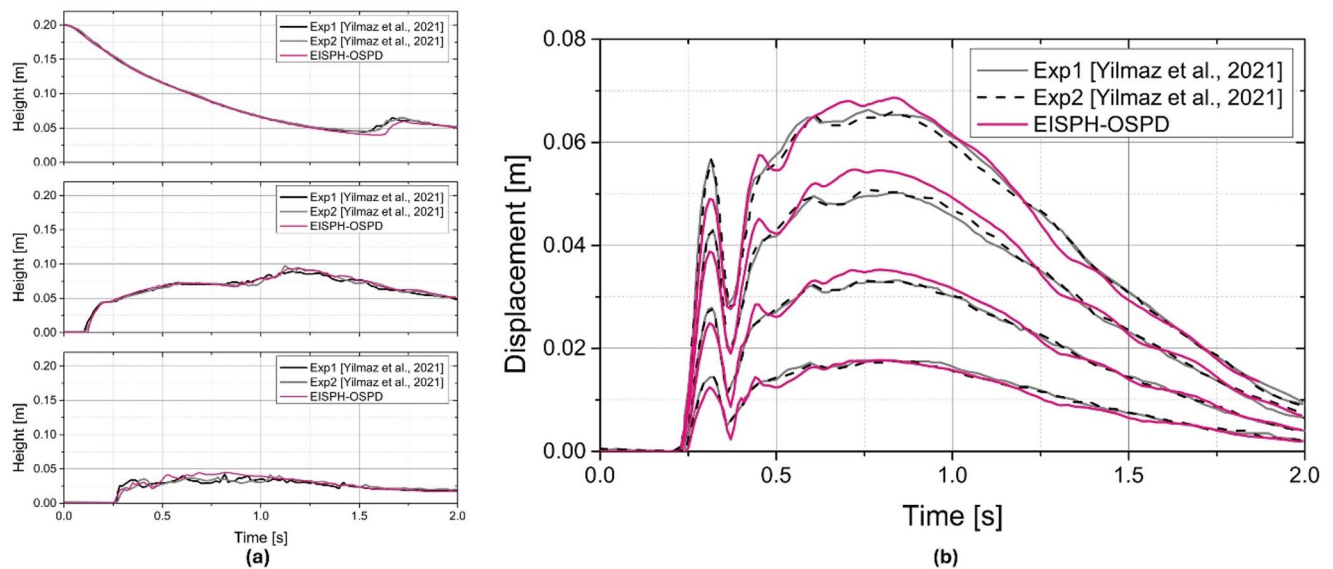


Fig. 29 Water Surface Elevation (H1–H3) and Gate Displacement (P1–P4) Compared with Experimental Results

modeled with a density $\rho^s = 2500 \text{ kg/m}^3$, Young's modulus $E^s = 1 \text{ MPa}$, and Poisson's ratio $\nu^s = 0$. The initial pressure distribution in the water column is prescribed using an asymptotic solution. Two particle resolutions are considered in this study: $\Delta x = a/5$ and $\Delta x = a/10$, where a is the thickness of the elastic gate displayed in Fig. 23c.

4.3.3.2 Numerical results As illustrated in Fig. 30, the proposed method accurately captures the sharp dynamics of the fluid interface and the corresponding structural deformation

of the elastic baffle, while maintaining a smooth and numerically stable pressure field. In the visualization, SPH fluid particles are colored by the normalized pressure, whereas PD structural particles are colored by dilatation, effectively highlighting the fluid–structure interaction behavior. In addition, Fig. 31 presents the time-evolving reconstructed stress field during the hydro-elastic impact, which exhibits smooth, physically consistent distribution and aids inter-

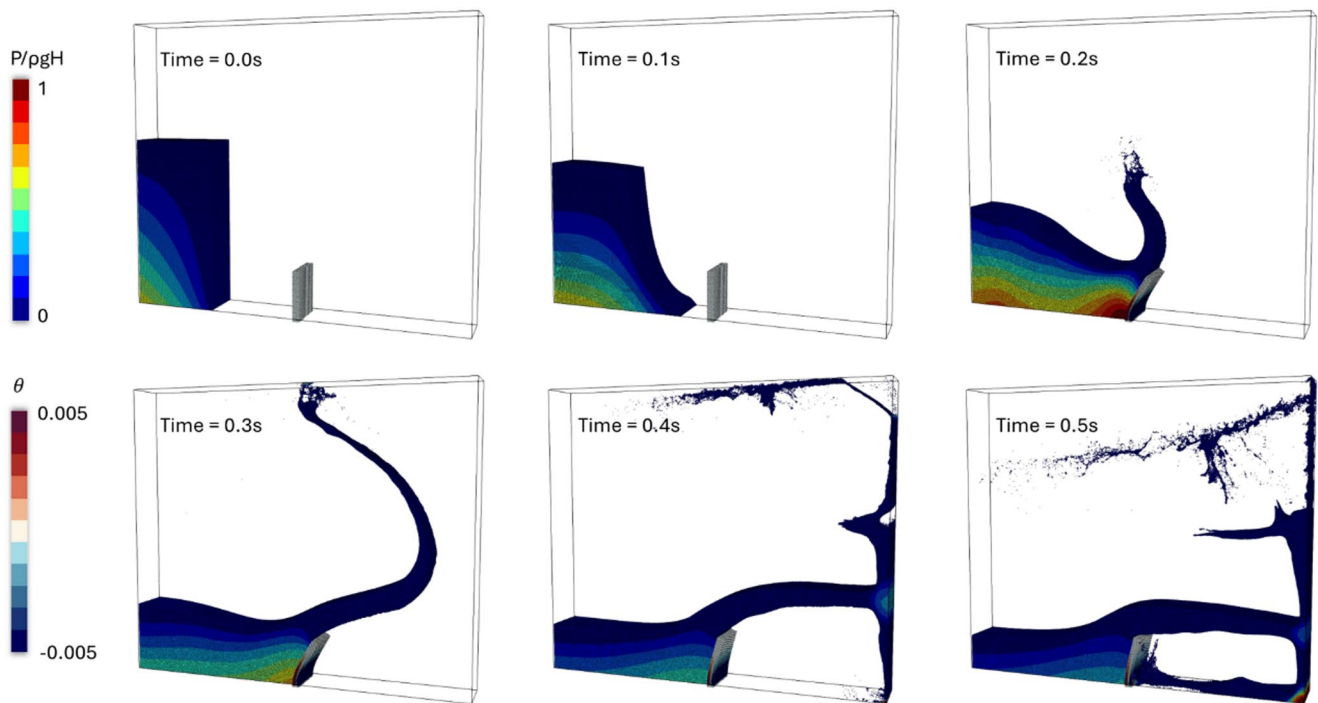


Fig. 30 Time Snapshots of Pressure Field of SPH Fluid Particles and Dilatation Fields of PD Structure Particles in Flow–Baffle Interaction Problem ($\Delta x = a/10$)

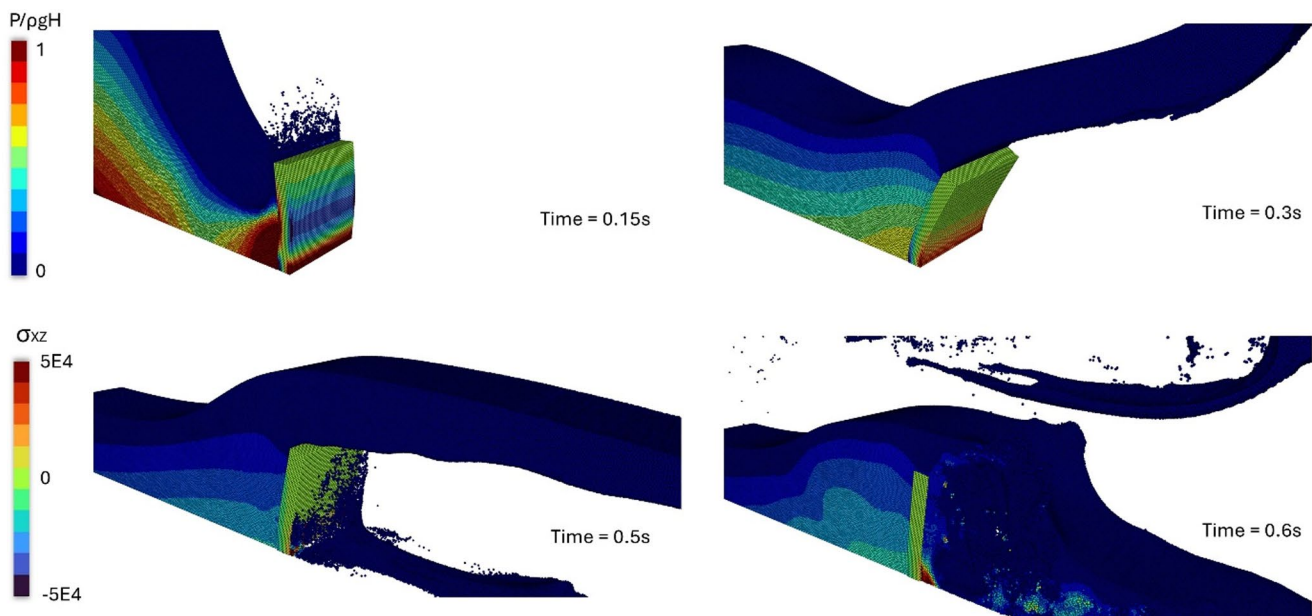


Fig. 31 Time Snapshots of Reconstructed Stress Field of PD Structure Particles in the Flow–Baffle Interaction ($\Delta x = a/10$)

pretation of local stress concentrations during the large deformation.

As the water column progresses and impacts the elastic baffle, the deformation of the structure induces vertical motion and wave-like behavior in the fluid. The elasticity of the baffle allows for realistic large-deformation responses,

which, in turn, lead to the formation of complex free-surface structures. Despite the complexity of the interface evolution, the pressure field remains smooth and continuous, demonstrating the stability and robustness of the proposed coupled numerical framework.

The deformation of the elastic baffle is quantitatively assessed by comparing the horizontal and vertical

displacements of its tip with results from other numerical studies conducted by different numerical methods involving WSPH-PD [21], SPEM [86], FEM [87], Particle FEM (PFEM) [88], and WSPH-SPH [89]. As shown in Fig. 32, for both coarse and fine particle resolutions, the tip displacements of the elastic baffle show good agreement with each other and with the reference results. The present results fall within the acceptable range of existing studies, demonstrating the reliability and accuracy of the proposed method.

4.3.4 Dam-break flow interacting with a fracturable elastic baffle attached to the bottom wall

4.3.4.1 Experimental condition Building upon the previous section, the current study introduces the assumption that the elastic baffle is fracturable. Since no corresponding experimental data are available for this scenario, the results are evaluated through comparison with other numerical methods. While the definition of the critical stretch s_C is differently defined among existing studies [21, 39], and its value should ideally be based on both numerical con-

ditions and experimental observations, a quantitative validation is mainly conducted using $s_C = 0.15$, following precedent in the literature [39]. Initial particle distance is set as $\Delta x = a/10$. To prevent artificial penetration between the PD baffle and adjacent solid boundaries, a repulsive force based on the Lennard-Jones potential is added to the momentum equation of PD particles, where the coefficient D is defined as 1.

$$F_{ij}^s = \begin{cases} D \left[\left(\frac{r_0}{r_{ij}} \right)^{12} - \left(\frac{r_0}{r_{ij}} \right)^4 \right] \frac{r_{ij}}{r_{ij}^2}, & r_{ij} \leq r_0 \\ 0, & \text{otherwise} \end{cases} \quad (58)$$

4.3.4.2 Numerical results Figure 33 presents the pressure field of the fluid and the dilatation field of the structure. As the hydrodynamic pressure exerts loading on the elastic baffle, the structure undergoes elastic deformation. Once the deformation reaches the critical stretch threshold, the brittle fracture initiates in the baffle. As shown in the figure, despite the occurrence of fracture, the pressure field remains

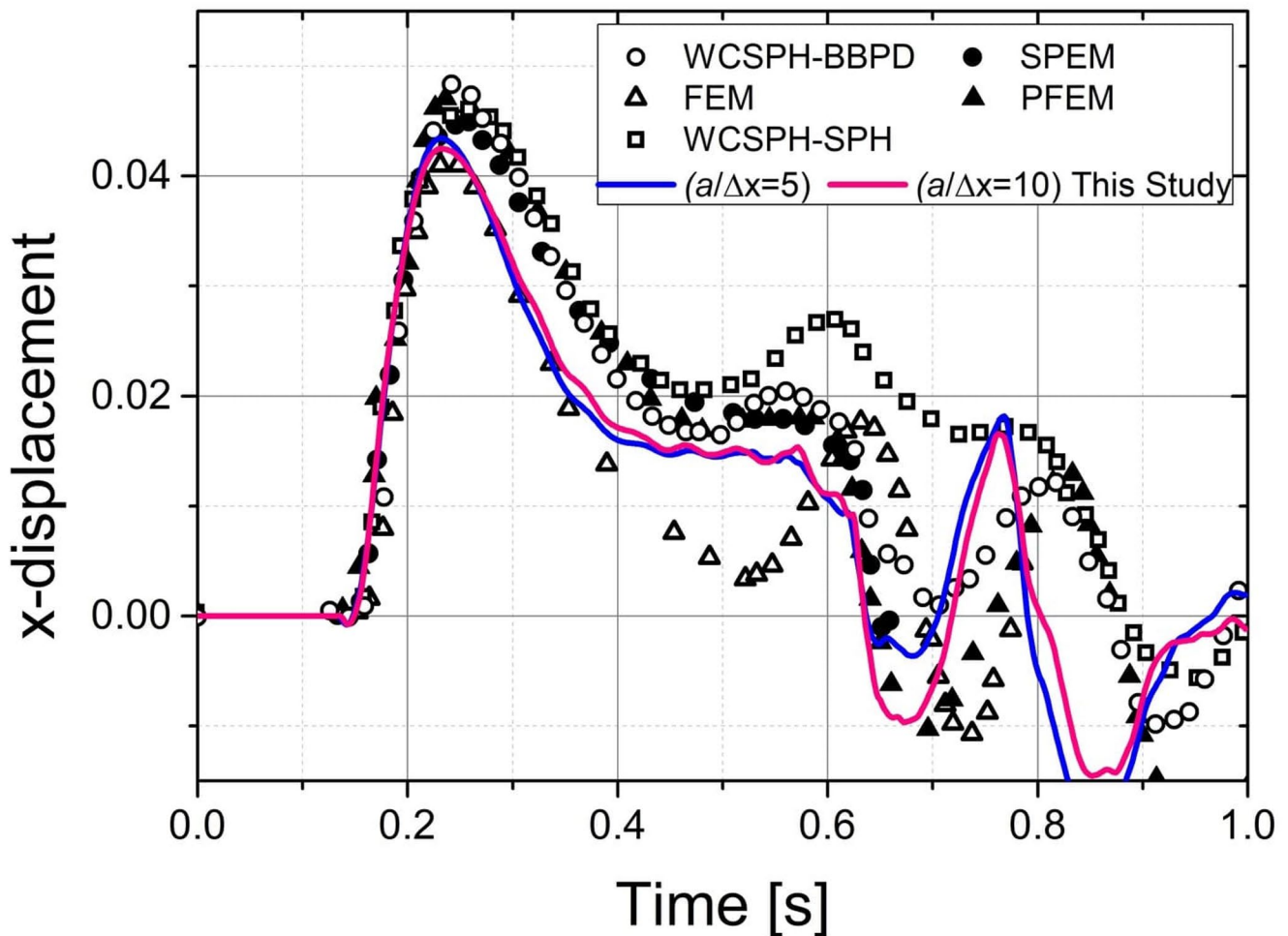


Fig. 32 Horizontal and Vertical Tip Displacements of Elastic Baffle Compared With Other Numerical Studies

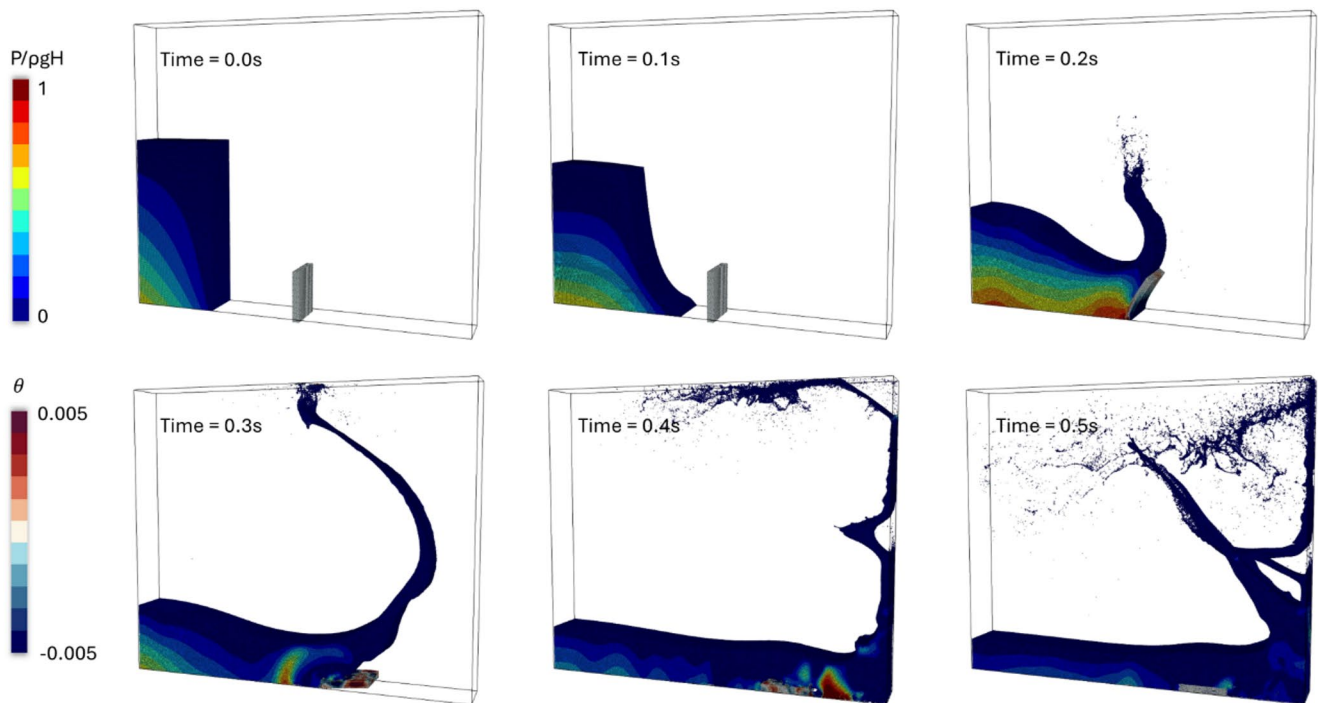


Fig. 33 Time Snapshots of Pressure Field of SPH Fluid Particles and Dilatation Fields of PD Structure Particles in Flow–Baffle Interaction Problem with Baffle Failure ($\Delta x = a/10$)

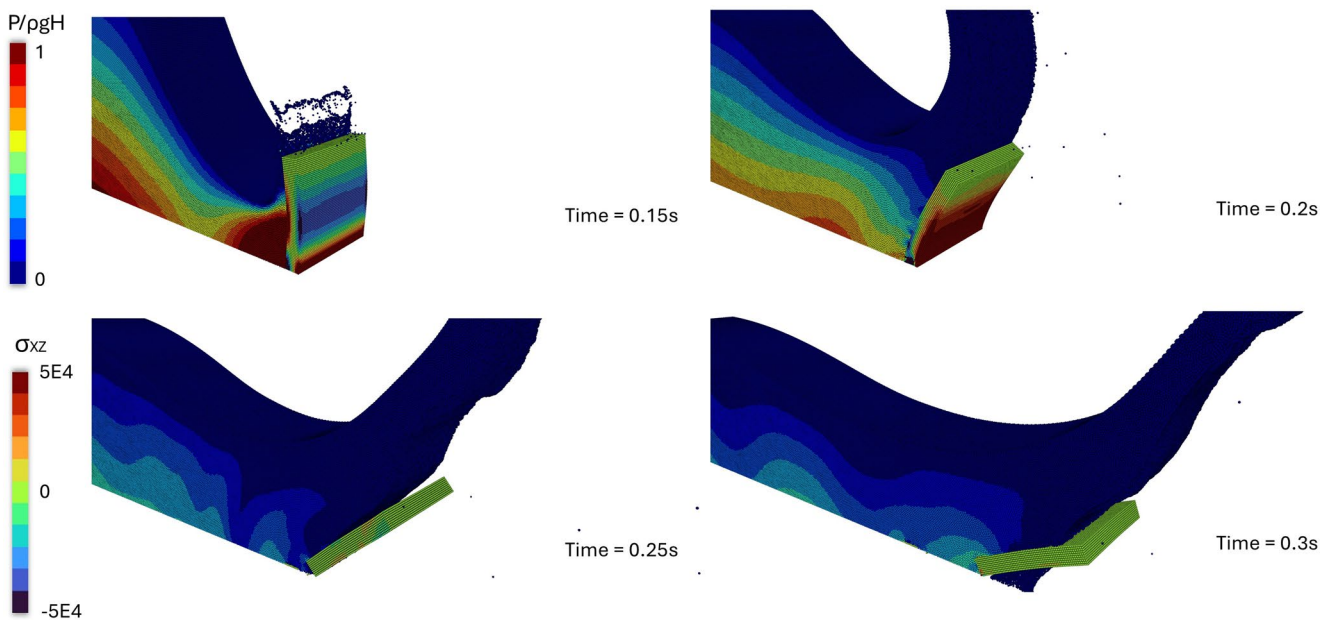


Fig. 34 Time Snapshots of Reconstructed Stress Field of PD Structure Particles in the Flow–Baffle Interaction with Baffle Failure ($\Delta x = a/10$)

smooth and continuous, exhibiting only minor oscillations. After the onset of fracture, pressure oscillations become more noticeable; however, they are progressively damped by viscous effects as the pressure wave propagates through the domain. Furthermore, Fig. 34 displays the reconstructed

stress field during the failure process, which remains smooth and continuous prior to crack initiation.

In Fig. 35, the time evolution of the z-displacement of the center of mass of the elastic baffle is presented, along with numerical results from previous studies employing different numerical methods, including SPH-FPM [90],

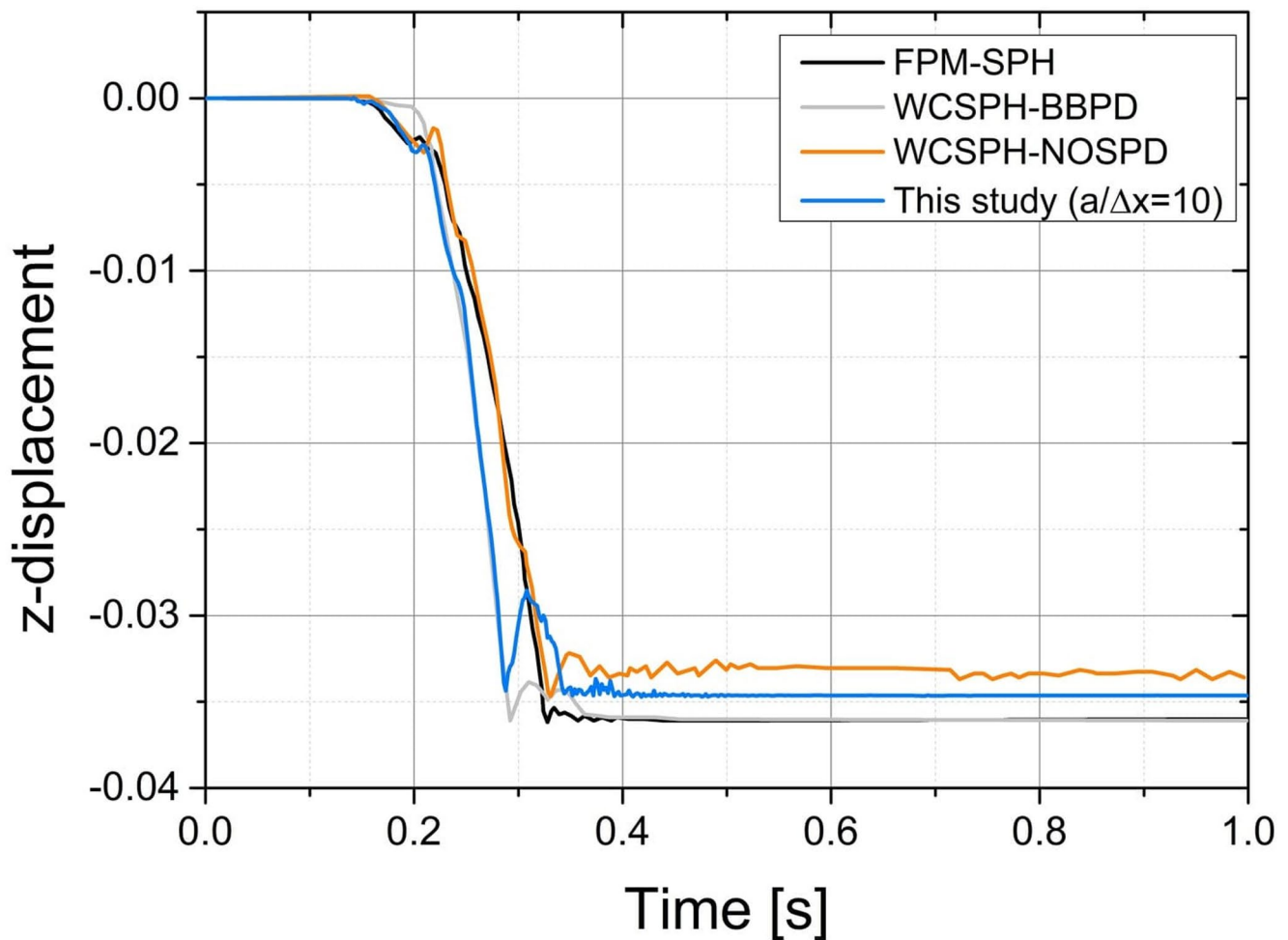


Fig. 35 Comparison of the Time Evolution of the Z-Displacement of the Center of Mass of the Elastic Baffle with Results from Other Numerical Studies

WCSPH-BBPD [21], WCSPH-NOSPD [39]. After a brief period of elastic deformation, the slope of the z-displacement evolution sharply increases, indicating the initiation of brittle fracture. All comparative studies exhibit fracture initiation at similar points. Following fracture, due to the significantly higher density of the elastic baffle which is more than twice that of water, and the early onset of fracture, the hydrodynamic pressure load does not induce substantial further movement of the baffle and is rapidly damped.

5 Conclusion

The present study proposed and validated a coupled EISPH-OSPD framework for fluid–structure interaction, focusing on accurately capturing the elastic response of solids and their interaction with incompressible flows. The fluid phase, modeled using the EISPH method, is coupled with a structural model based on OSPD through momentum exchange and subcycling-based time integration management.

Comprehensive benchmark tests, including both standalone and coupled simulations, demonstrated the robustness and accuracy of the proposed framework across a range of scenarios, from elastic deformation to brittle fracture.

Two key aspects were emphasized. First, the elastic behavior of the OSPD model, which has received limited attention in prior work, was rigorously evaluated. Surface correction was found to be essential for accurate stress representation, significantly outperforming volume correction regardless of the selected influence function. Second, the time integration scheme played a critical role: the first-order Euler method led to nonphysical oscillations, whereas the second-order predictor–corrector scheme yielded stable and accurate results.

The coupled method was then applied to a series of fluid–structure interaction benchmark problems. The hydrodynamic pressure field remained highly stable, and the resulting pressure loads induced structural deformation in a consistent and physically plausible manner. The dilatation field also remained stable across various fluid–structure

impact scenarios. The proposed framework showed strong agreement with both experimental observations and numerical results reported in the literature. Notably, the incompressible SPH method contributed to pressure stability without requiring any additional stabilization terms, even in simulations involving structural failure, demonstrating its effectiveness for complex fluid–structure interaction problems.

Overall, the developed framework offers a physically consistent, stable, and accurate approach for simulating fluid–structure interaction problems involving large elastic deformations and fracture. In this study, OSPD was employed in its linear elastic form, which has been widely applied to model damage and failure up to moderately large deformations. Although nonlinear models such as hyperelastic/plastic or hypoelastic/plastic formulations may provide more accurate representations under extreme strain conditions, the present approach provides sufficient fidelity for capturing the failure process within the targeted deformation range. Future improvements may involve incorporating nonlinear constitutive models to address more complex behaviors, as well as extending the framework to include adaptivity, multiphase flow modeling, and thermal interaction effects.

Acknowledgements Author Eung Soo Kim acknowledges support from the National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIT) (No. 2021M2D2A1A03046881).

Funding Open Access funding enabled and organized by Seoul National University.

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